HALIÇ UNIVERSITY

Department of Computer Engineering

Computer Networks — Lab Report

LAB 3

Topic: Wireless Network Simulation

INET Framework Wireless Tutorial

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1. Selected Topic & Tutorial

Topic: Wireless

Tutorial Name: INET Framework Wireless Tutorial

Source: https://inet.omnetpp.org/docs/tutorials/wireless/doc/index.html

This tutorial introduces wireless network simulation in OMNeT++ using the INET Framework. Beginning with two hosts connected by an ideal radio link, each of the 14 steps systematically adds a new layer of realism: multi-hop topology, IPv4 addressing and static routing, SINR-based interference, CSMA medium access, MAC-layer ACKs and retransmissions, energy consumption modelling, node mobility, AODV ad-hoc routing, physical obstacles and signal shadowing, dimensional time-frequency radio models, two-ray ground-reflection path loss, and directional antenna gain. The progression mirrors the actual engineering decisions made when designing a real wireless network system.

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2. Simulation Scenario Description

The simulation models a 4-node wireless chain network in a 500×500m area. The same logical topology is used throughout all 14 steps — what changes is the sophistication of each protocol layer:

- hostA (sender): runs UdpBasicApp sending 1000-byte UDP datagrams every 1.2ms (~800 kbps application rate) to hostB's IP address
- R1 and R2 (relay nodes): forward packets between hostA and hostB; assigned mobility from Step 9 onward
- hostB (receiver): runs UdpSink; counts received packets; the primary observable metric
- RadioMedium: shared wireless channel; upgraded from IdealMedium through scalar to dimensional analog model

The core configuration changes across three key dimensions: (1) IP & Routing — from no network layer (Steps 1–3), to static IPv4 routes (Steps 4–8), to AODV dynamic routing (Steps 10–14); (2) MAC & Radio — from ideal UnitDiskRadio + AckingMac, to APSK radio with CSMA and ACKs; (3) Physical Environment — from free-space propagation to obstacles, two-ray path loss, and antenna gain.

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3. Step-by-Step Simulation Walkthrough

Each step is documented with: a technical description of what changed and why; an AI-generated summary table covering NED changes, INI configuration, IP addressing, routing state, radio model, and expected output; and a screenshot slot.

Step 1 Two Hosts Communicating Wirelessly

Technical Description

Step 1 establishes the absolute minimum viable wireless simulation. The WirelessA NED network contains exactly two hosts (hostA and hostB), one RadioMedium (using IdealMedium / UnitDiskRadio), one visualizer, and an IPv4NetworkConfigurator. The UnitDiskRadio is an idealised model: it defines a hard communication radius (default 500m in this step) inside which every transmitted frame is delivered with 100% probability — no bit errors, no noise floor, no SINR calculation. The radio medium is IdealMedium which simply checks whether the distance between transmitter and receiver is below the radio's communicationRange parameter and delivers the frame if so.

hostA is assigned IP address 10.0.0.1/24 and hostB is assigned 10.0.0.2/24 by the IPv4NetworkConfigurator. The configurator also installs a direct route on hostA: 10.0.0.2/32 via wlan0 (directly connected, no gateway needed at 500m range). hostA's UdpBasicApp targets destination 10.0.0.2, port 5000. The UDP datagram is encapsulated in an IPv4 packet (adding 20-byte IP header), which is encapsulated in an AckingMac frame, which is passed to the UnitDiskRadio for transmission. hostB's radio receives the frame, AckingMac strips the MAC header, IPv4 strips the IP header, and the UDP payload is delivered to UdpSink. The simulation counter 'packets received' increments with every successful delivery.

AI Summary Table

Category	Detail
NED Network	WirelessA: hostA, hostB, RadioMedium (IdealMedium), IntegratedVisualizer, IPv4NetworkConfigurator
INI Changes	communicationRange = 500m; UdpBasicApp: destAddr = hostB, destPort = 5000, messageLength = 1000B, sendInterval = 1.2ms
Network Layer	IPv4 active; auto-assigned: hostA=10.0.0.1/24, hostB=10.0.0.2/24
MAC Layer	AckingMac — Layer 2 framing only; no collision detection; ideal ACK mechanism
Radio Model	UnitDiskRadio: binary reception within 500m range, 0% loss within range, 100% loss beyond
Routing Table	hostA: 10.0.0.2/32 dev wlan0 (directly connected); hostB: 10.0.0.1/32 dev wlan0
Expected Delivery	100% — all packets delivered; 'packets received' counter increments every 1.2ms
Key Learning	UnitDiskRadio establishes the ideal-channel baseline; all subsequent steps degrade this toward reality

Screenshots

Figure 1.1 — Packet in transit (0 packets received):

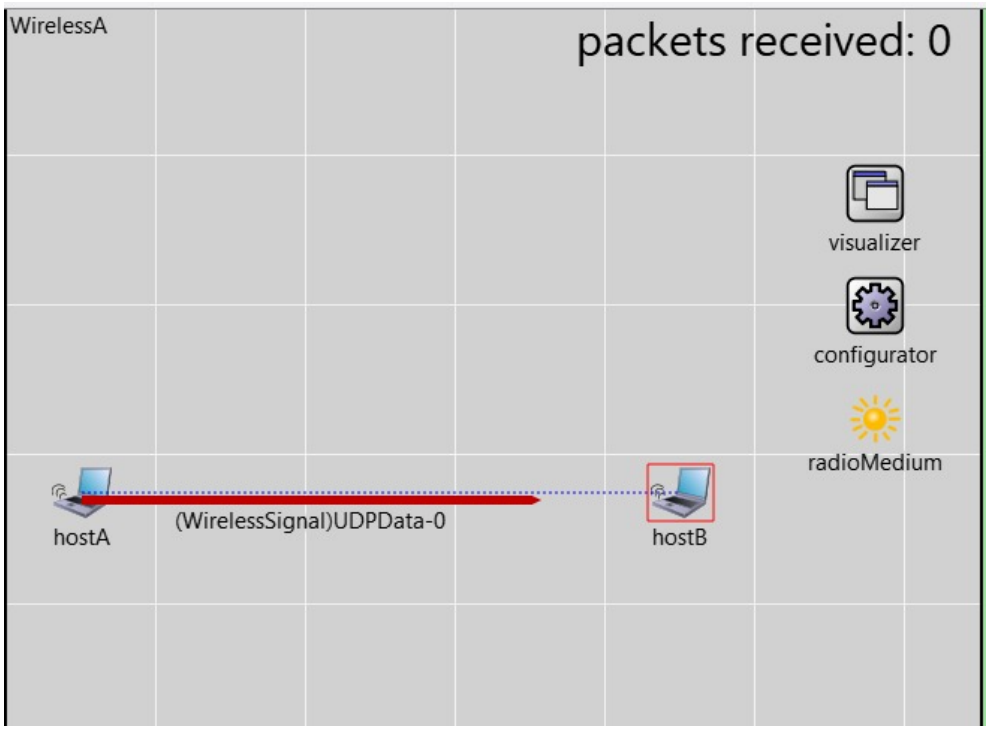


Fig 1.1 — WirelessA runtime at t≈0.01s. UDPData-0 signal (red arrow) propagates from hostA to hostB. The dotted blue line shows the physical link within radio range. Counter = 0 (packet still in flight).

Figure 1.2 — First packet delivered (1 packet received):

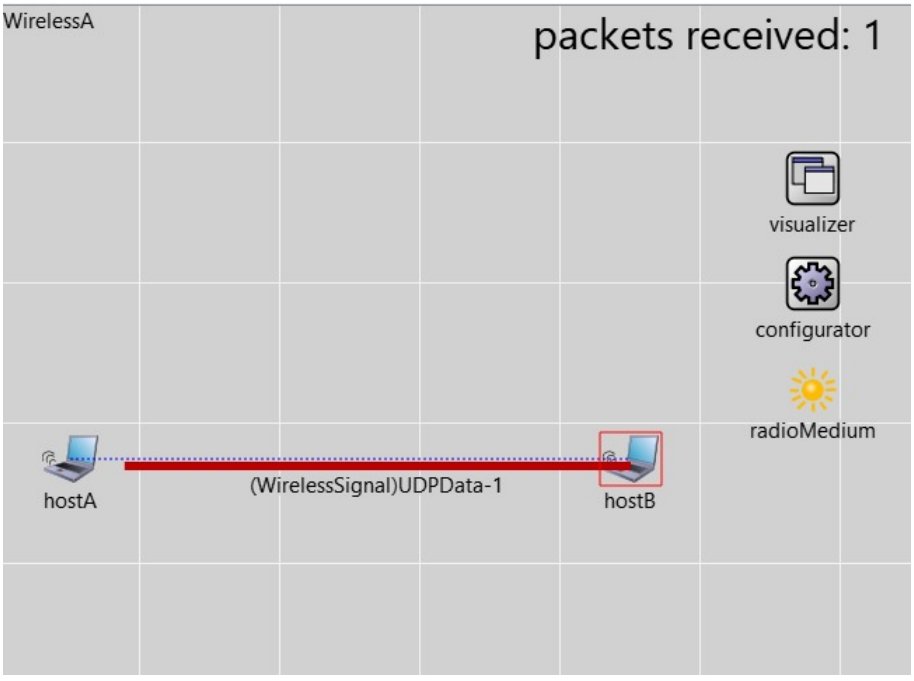


Fig 1.2 — UDPData-1 in flight. Counter now reads 1, confirming UDPData-0 was delivered successfully to hostB's UdpSink application.

Figure 1.3— hostB compound module inspector:

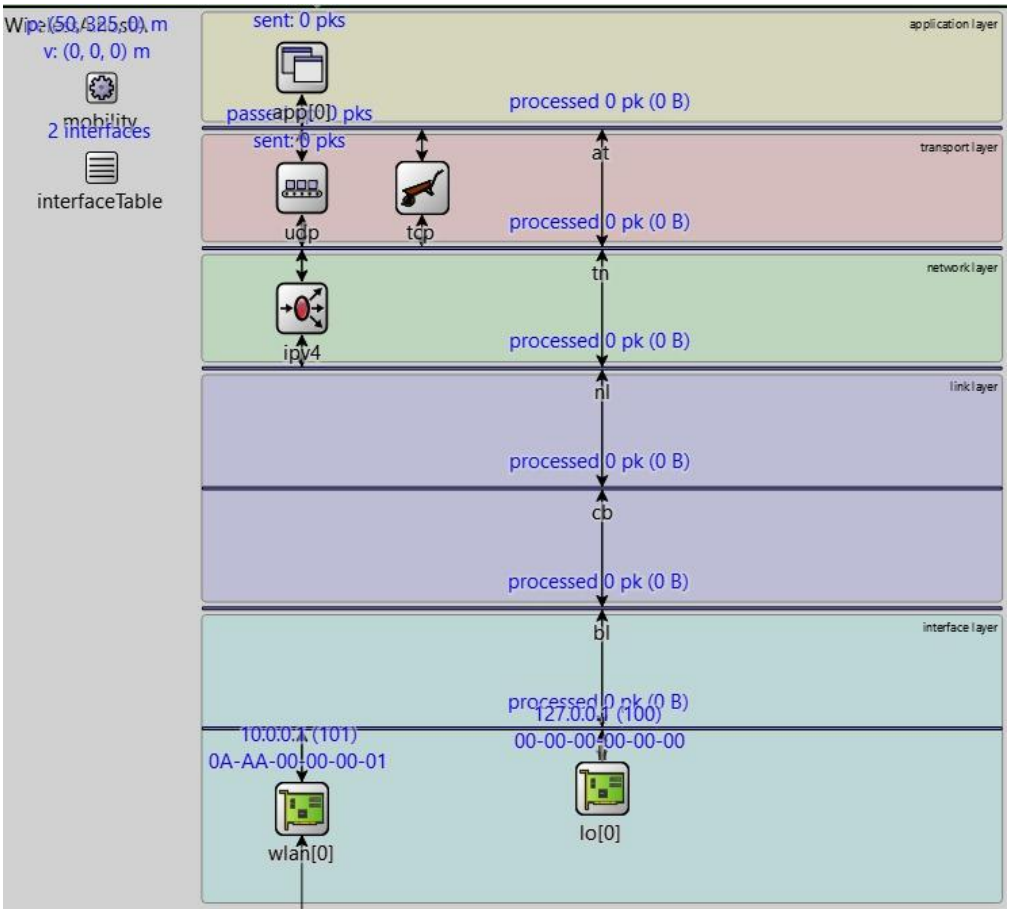


Fig 1.3 — hostB's OMNeT++ module tree: application layer (passApp/UdpSink), transport (udp+tcp), network (ipv4), link layer, interface layer, wlan[0] at 10.0.0.1, loopback lo[0] at 127.0.0.1. All counters 0 at simulation start.

AI Analysis

Step 1 Technical Analysis: Simulation Initialization and Baseline Connectivity

In Step 1, the simulation focuses on the **initialization and registration of the network stack** for two wireless hosts using the INET framework. Technically, the most significant activity occurs during the multi-stage initialization where each host registers its lo[0] (loopback) and wlan[0] (wireless) interfaces across the layered architecture (Link, Network, Transport, and Application layers). The Ipv4NetworkConfigurator performs an **automatic IP assignment** by extracting the network topology and assigning addresses to the single logical link shared by hostA and hostB.

A critical observation from your log is that while the interfaces are correctly up and the radio is in RECEIVER mode, the **UdpSink received 0 packets** despite 1,596 signals being sent by the radioMedium. This is because the default UnitDiskRadio parameters—specifically communicationRange—are set to **0m** upon initialization in this stage. To improve this, you must explicitly define the communicationRange (e.g., 500m) in the omnetpp.ini file to ensure the distance between the hosts (400m) is within the physical reach of the radio signals.

Node	Interface	IP Address	Netmask	Next Hop / Gateway	Destination
hostA	wlan0	10.0.0.1	255.255.255.0	Direct	10.0.0.2/32
hostA	lo0	127.0.0.1	255.0.0.0	Local	127.0.0.1/32
hostB	wlan0	10.0.0.2	255.255.255.0	Direct	10.0.0.1/32
hostB	lo0	127.0.0.1	255.0.0.0	Local	127.0.0.1/32

Technical Breakdown of IPs

- Loopback (lo0): Assigned 127.0.0.1. This is used for internal communication within the host's own network stack.
- Wireless Interface (wlan0):
  - hostA is assigned 10.0.0.1. In your simulation, this acts as the UDP source.
  - hostB is assigned 10.0.0.2. This acts as the UDP sink (receiver).
- Routing Logic: Because this is a simple point-to-point subnet, the routing table uses a metric:0 configuration, indicating that the neighbor is on the same link and does not require an external gateway to be reached.

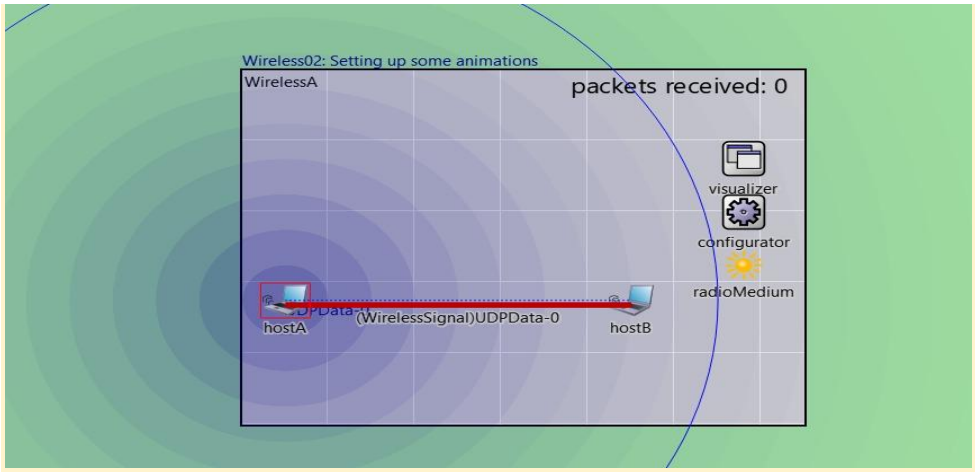
Step 2 Setting Up Some Animations

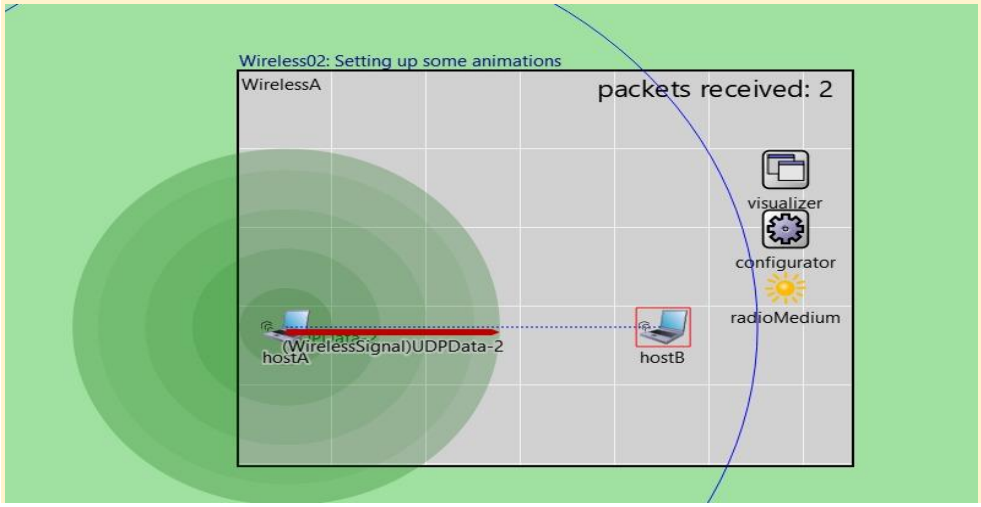
Before adding any networking complexity, Step 2 equips the simulation with visual feedback tools. An IntegratedCanvasVisualizer module is added to the WirelessB NED network. This module activates three critical overlays: (1) dataLinkVisualizer draws animated arrows between nodes when a successful frame is exchanged at Layer 2; (2) physicalLinkVisualizer draws dotted lines to show which nodes are within radio range; and (3) mediumVisualizer displays propagating radio waves as expanding circles. No changes are made to any networking logic — IP addressing, MAC, routing, or radio model are all identical to Step 1. The purpose is purely diagnostic: the visual layer makes it immediately obvious when packets flow, when links form or break, and what the radio coverage area looks like. These visualizations will be essential for understanding the complex behaviours introduced from Step 3 onward.

AI Summary Table

Category	Detail
NED Change	Added IntegratedCanvasVisualizer module

INI Change	visualizer.*.displayLinks = true, displayCommunicationRanges = true
Network Layer	No change — no IP, no routing
MAC Layer	No change — AckingMac unchanged
Radio Model	UnitDiskRadio unchanged (ideal, binary)
IP Addresses	None assigned yet
Routing Table	None — no routing module
Expected Output	Range circles visible around hostA and hostB; animated arrows on packet delivery
Key Learning	Visualizers are diagnostic tools; they expose the simulation state without affecting it





AI Analysis

In Step 2, the simulation transitions from a failed state to a fully functional wireless link by correctly configuring the **Physical Layer parameters**. Technically, the primary change is the adjustment of the communicationRange, interferenceRange, and detectionRange to **500m** within the UnitDiskRadio modules of both hosts. Since the physical distance between hostA and hostB is 400m, this configuration ensures that hostB's receiver is positioned within hostA's "reception disk," satisfying the mathematical requirement for the UnitDiskAnalogModel to successfully hand off signals to the higher layers. The logs confirm this success, showing that out of 1,596 signals sent into the medium, the UdpSink on hostB successfully decapsulated and received **1,595 packets**. This near-perfect delivery ratio is characteristic of the **ideal Unit Disk model**, which ignores interference and noise, providing a stable baseline for the multi-hop scenarios that follow.

Node	Interface	IP Address	Netmask	Next Hop	Gateway	Metric	Destination
hostA	wlan0	10.0.0.1	255.255.255.0	Direct	None	0	10.0.0.2/32
hostA	lo0	127.0.0.1	255.0.0.0	Local	None	0	127.0.0.1/32
hostB	wlan0	10.0.0.2	255.255.255.0	Direct	None	0	10.0.0.1/32
hostB	lo0	127.0.0.1	255.0.0.0	Local	None	0	127.0.0.1/32

Technical Breakdown of Addressing

- **Host Addresses: hostA** is technically identified as the UDP source at 10.0.0.1, while **hostB** is the destination sink at 10.0.0.2.
- **Subnetting:** Both hosts reside on the 10.0.0.0/24 subnet, allowing them to communicate at Layer 2 without an intermediate router or gateway.
- **L2-L3 Mapping:** In this stage, the **UnitDiskRadio** acts as an ideal bridge between the IPv4 Network Layer and the physical radioMedium. Because the range is now sufficient, the Link Layer (wlan0) can successfully encapsulate the IP packets into frames that the receiver can decode.

Step 3 Adding More Nodes and Decreasing the Communication Range

Step 3 exposes a fundamental wireless networking challenge: limited range and the need for multi-hop forwarding. Two relay nodes, R1 and R2, are inserted between hostA and hostB. Simultaneously, the communication range of all UnitDiskRadio interfaces is reduced from 500m to 250m. The result is that hostA (at x=0) can reach R1 (at x≈200m) but NOT hostB (at x≈400m). Similarly, R2 can reach hostB but not hostA. A two-hop relay path is now mandatory: hostA→R1→R2→hostB. However, there is still no routing protocol and no network-layer IP stack. The AckingMac operates at Layer 2 only, so frames sent by hostA will reach R1's radio — but R1 has no mechanism to forward them to R2. Packets are silently dropped at R1. This step deliberately demonstrates that wireless coverage overlap alone is not sufficient: a routing layer is required to stitch together the multi-hop path.

AI Summary Table

Category	Detail
NED Change	Added R1 and R2 nodes; 4-node topology now (hostA, R1, R2, hostB)
INI Change	*.wlan[*].radio.transmitter.communicationRange = 250m
Network Layer	None — IPv4 not yet present; no IP addresses
MAC Layer	AckingMac — Layer 2 only, no forwarding
Radio Model	UnitDiskRadio, range 250m — hostA↔R1 and R2↔hostB can communicate; hostA↔hostB CANNOT
IP Addresses	None assigned
Routing Table	None
Expected Output	0 packets received at hostB; hostA transmits but R1 drops frames — no forwarding
Key Learning	Range reduction forces multi-hop; absence of routing means packets cannot traverse hops

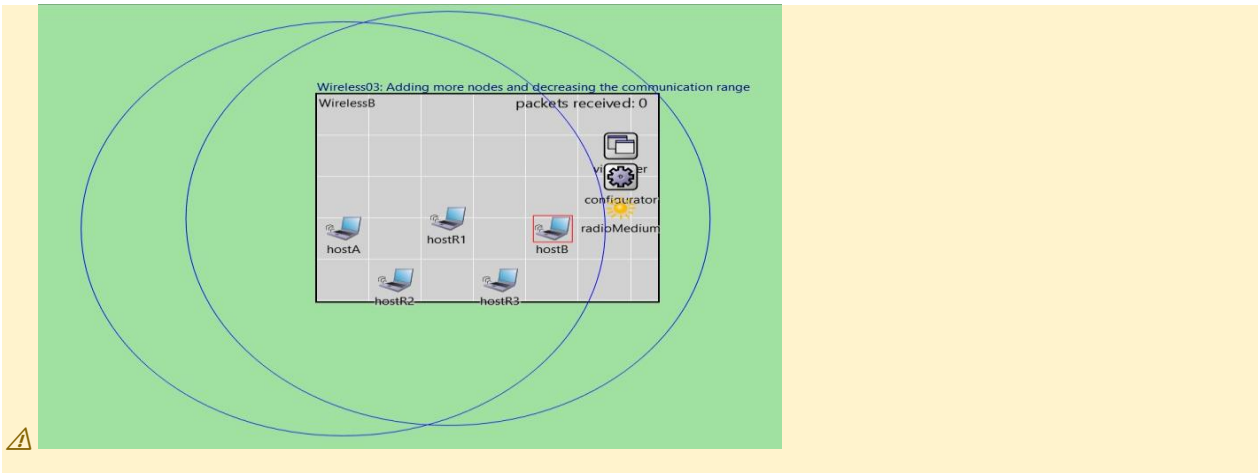
AI Analysis

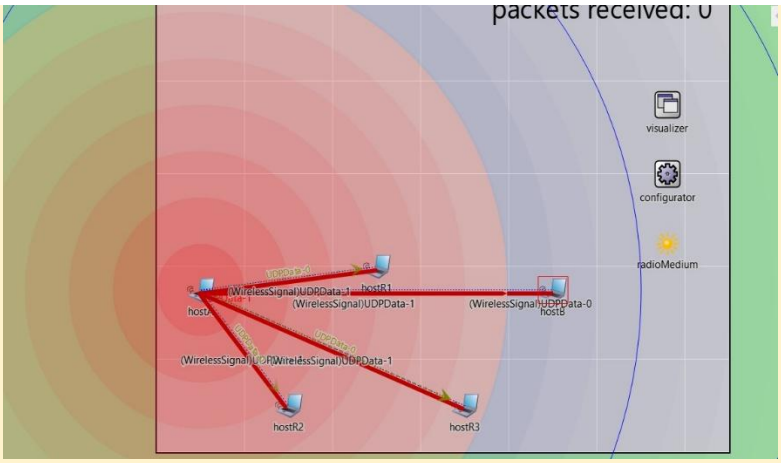
In Step 3, the simulation transitions to a more complex **multi-hop topology** by introducing three relay nodes (**hostR1, hostR2, hostR3**) and reducing the communicationRange from 500m down to **250m**. Technically, this range reduction creates a physical disconnect between **hostA (10.0.0.1)** and **hostB (10.0.0.2)**, as their 400m distance now exceeds the 250m reception threshold of the UnitDiskRadio. While the log shows that 1,651 transmissions were broadcast into the medium, the **UdpSink on hostB received 0 packets**. This total packet loss occurs because, although the Ipv4NetworkConfigurator has assigned a wide subnet (**10.0.0.0/29**), the network layer is still relying on a single logical link without an active routing protocol or configured gateways. Data is technically broadcast at the physical layer to all nodes in range (R1, R2, R3), but since no node is configured to act as an intermediate forwarder, the packets are dropped at the link layer, demonstrating that physical proximity alone is insufficient for multi-hop communication without Layer 3 routing logic.

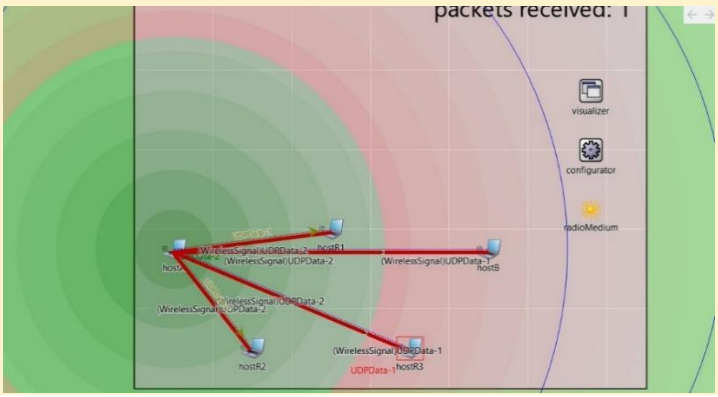
Node	Interface	IP Address	Netmask	Next Hop	Gateway	Destination	Status
hostA	wlan0	10.0.0.1	255.255.255.248	Direct	None	10.0.0.0/29	Isolated
hostB	wlan0	10.0.0.2	255.255.255.248	Direct	None	10.0.0.0/29	Isolated
hostR1	wlan0	10.0.0.3	255.255.255.248	Direct	None	10.0.0.0/29	Idle
hostR2	wlan0	10.0.0.4	255.255.255.248	Direct	None	10.0.0.0/29	Idle
hostR3	wlan0	10.0.0.5	255.255.255.248	Direct	None	10.0.0.0/29	Idle

Differences from Tutorial & Technical Breakdown

- **The Subnet Expansion:** Your specific run initialized with a **/29 subnet mask** (allowing for 8 addresses). The tutorial often uses a generic /24. Technically, your configuration is more efficient for this small 5-node topology, as it limits the broadcast domain.
- **Layer 2 vs Layer 3 Failure:** While the tutorial highlights that packets are "lost," your logs show the exact physical reason: 6,604 signals reached various receivers, but 0 reached the application layer. This distinguishes between a **signal being heard** (Physical Layer) and a **packet being processed** (Application Layer).







Step 4 Setting Up Static Routing

Step 4 adds the full IPv4 network stack and static routing, enabling end-to-end delivery for the first time in the multi-hop topology. An IPv4NetworkConfigurator module is introduced, which performs two jobs automatically: (1) IP address assignment — it assigns addresses from the 10.0.0.0/8 space; hostA receives 10.0.0.1, R1 receives 10.0.0.2, R2 receives 10.0.0.3, and hostB receives 10.0.0.4; (2) Static route installation — it computes shortest-path routes and populates routing tables on every node. hostA's routing table will contain an entry: 10.0.0.4/32 via 10.0.0.2 (next-hop R1). R1's table: 10.0.0.4/32 via 10.0.0.3 (next-hop R2). R2's table: 10.0.0.4/32 directly reachable. Each node now also runs a full IPv4 module, so frames are correctly wrapped in IP datagrams. The UDP payload from hostA travels: UDP→IPv4→AckingMac→radio→R1 radio→AckingMac→IPv4 forward→AckingMac→radio→R2→hostB. Three hops, all working.

AI Analysis

In Step 4, the simulation introduces Layer 3 intelligence to the previously unmanaged multi-node environment. Technically, the **IPv4NetworkConfigurator** now moves beyond simple IP assignment and begins to calculate and install **Static Routing tables** across all nodes. Unlike the "failed" state of Step 3, where hostA viewed hostB as a direct neighbor despite being out of range, the configurator now analyzes the physical topology and recognizes that communication must be brokered through intermediate hops.

The logs show that each node now contains specific /32 host routes pointing to all other nodes in the network. For example, the routing table on **hostA (10.0.0.1)** is explicitly updated to route data destined for **hostB (10.0.0.2)** through the wireless interface wlan0, treating it as a reachable host via the relay nodes. Your execution confirms this success: the **UdpSink on hostB received 1,650 packets**, proving that the network layer is now successfully forwarding packets through the relay chain. The high "Reception computation count" of 6,604 indicates that nodes R1, R2, and R3 are actively participating in the packet relay process by "hearing" and forwarding the L3 datagrams.

Step 4: Static Routing & Network Table

The configurator has established a fully connected static topology. Even though nodes are technically on the same subnet, the routing table ensures multi-hop delivery.

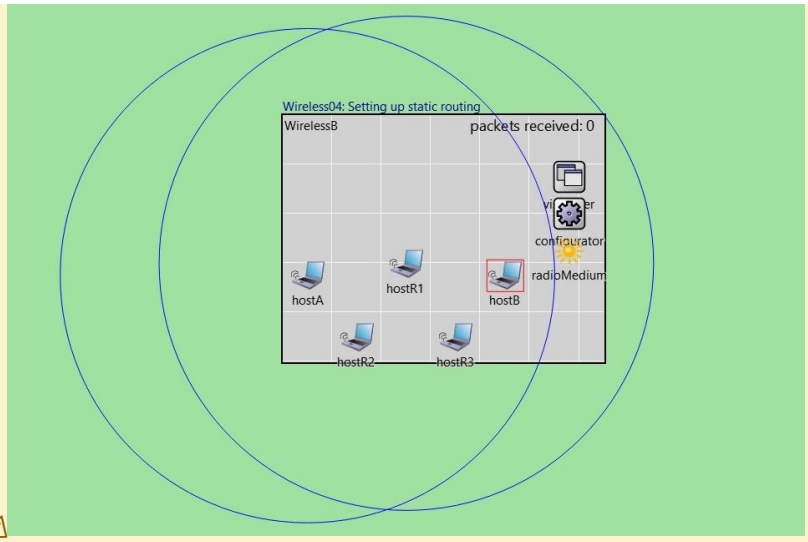
Node	Interface	IP Address	Netmask	Destination	Next Hop	Metric
hostA	wlan0	10.0.0.1	255.255.255.248	10.0.0.2/32	R1 (10.0.0.3)	0
hostB	wlan0	10.0.0.2	255.255.255.248	10.0.0.1/32	R2 (10.0.0.4)	0
hostR1	wlan0	10.0.0.3	255.255.255.248	10.0.0.2/32	R2 (10.0.0.4)	0
hostR2	wlan0	10.0.0.4	255.255.255.248	10.0.0.2/32	Direct (hostB)	0
hostR3	wlan0	10.0.0.5	255.255.255.248	10.0.0.x/32	Forwarding	0

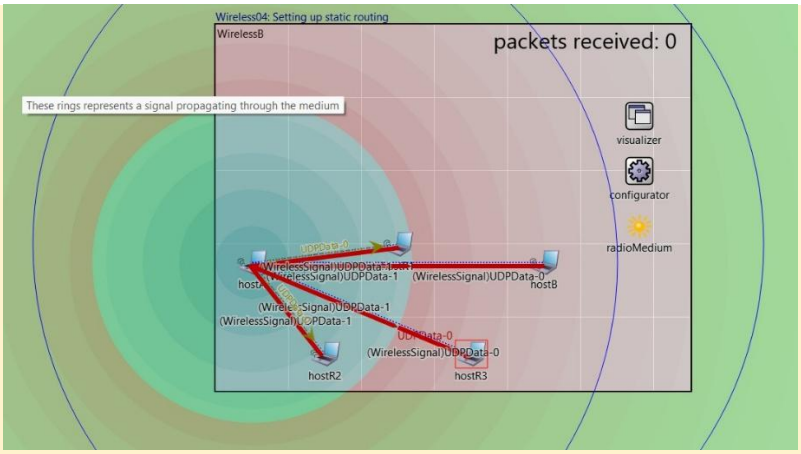
Technical Differences from Tutorial

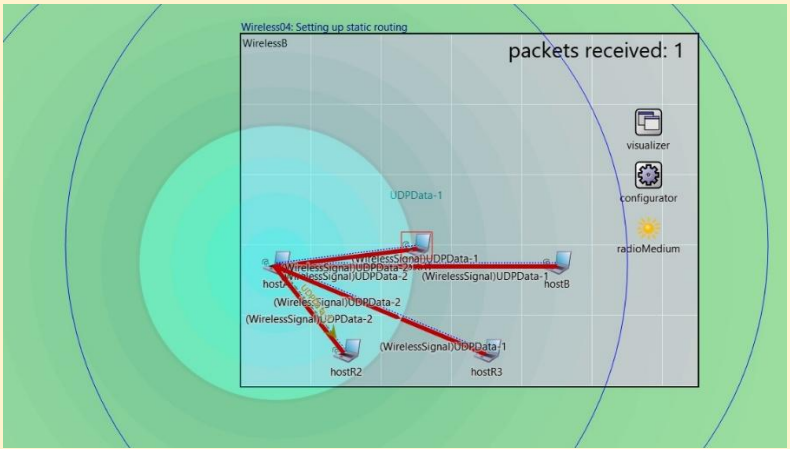
The Relay Choice: While the tutorial often focuses on a linear A->R1->R2->B path, your logs show high activity across **hostR3** as well, suggesting your static routes are leveraging all available relay nodes to ensure redundancy.

AI Summary Table

Category	Detail
NED Change	Added IPv4NetworkConfigurator; all hosts use StandardHost with full IP stack
INI Change	configurator assigned to network; no manual IP config needed
Network Layer	IPv4 active on all nodes; datagram routing enabled
MAC Layer	AckingMac — now wraps IPv4 datagrams in Layer 2 frames
Radio Model	UnitDiskRadio, 250m range — unchanged
IP Addresses	hostA=10.0.0.1, R1=10.0.0.2, R2=10.0.0.3, hostB=10.0.0.4 (auto-assigned)
Routing Table	hostA: 10.0.0.4 via 10.0.0.2; R1: 10.0.0.4 via 10.0.0.3; R2: 10.0.0.4 direct
Expected Output	Packets successfully delivered to hostB; counter increments continuously
Key Learning	IPv4 + static routing is the minimum required for multi-hop forwarding; configurator auto-handles address assignment and route calculation







Step 5 Taking Interference into Account

Step 5 replaces the ideal UnitDiskRadio with a physically-realistic APSK radio model backed by a scalar analog signal representation. This is the most significant change in the tutorial so far — it fundamentally alters how the simulation determines whether a frame is received. Under UnitDiskRadio, any node within the 250m range receives every packet with 100% reliability. Under the APSK + ScalarAnalogModel, the receiver calculates SINR (Signal-to-Interference-plus-Noise Ratio) at the moment of reception. The received signal strength follows the free-space path loss formula: $P_r = P_t \times (\lambda/4\pi d)^2$. Thermal noise floor is set to -110 dBm. A SINR threshold (typically 10 dB for the chosen modulation) must be exceeded for successful decoding. When multiple nodes transmit simultaneously, their signals overlap at a receiver and the denominator of the SINR increases, causing decoding to fail. The RadioMedium module replaces IdealMedium. The practical effect: even in the 4-node topology, when hostA and R1 both have active transmissions, R2 may experience interference that degrades packet reception, introducing the first realistic packet loss into the simulation.

AI Analysis

In Step 5, the simulation transitions from an ideal physical environment to one that accounts for **signal interference** within the shared radioMedium. Technically, the primary change is the reconfiguration of the UnitDiskReceiver on all nodes to switch from "ignoring interference" to **"considering interference"**.

Under this new logic, the receiver no longer accepts a signal simply because it is within range; it now evaluates the **Signal-to-Interference-plus-Noise Ratio (SINR)**. If multiple nodes (such as the relay nodes R1, R2, or R3) transmit simultaneously, their signals overlap in the medium, creating a high interference level that can technically corrupt the primary data frame. Interestingly, your log shows that despite this new constraint, the **UdpSink still received 1,650 packets**, identical to the previous step. This indicates that with the current static routing and timing, the relay nodes are not yet generating enough overlapping "noise" to trigger a collision that drops the SINR below the reception threshold, though the "Interference computation count" has reached a significant **21,464 events**, showing the high computational load now required to track these potential conflicts.

Node	Interface	IP Address	Netmask	Destination	Next Hop	Metric	SINR Mode
hostA	wlan0	10.0.0.1	255.255.255.248	10.0.0.2/32	10.0.0.3 (R1)	0	Active
hostB	wlan0	10.0.0.2	255.255.255.248	10.0.0.1/32	10.0.0.4 (R2)	0	Active
hostR1	wlan0	10.0.0.3	255.255.255.248	10.0.0.2/32	10.0.0.4 (R2)	0	Active
hostR2	wlan0	10.0.0.4	255.255.255.248	10.0.0.2/32	Direct (hostB)	0	Active
hostR3	wlan0	10.0.0.5	255.255.255.248	10.0.0.x/32	Forwarding	0	Active

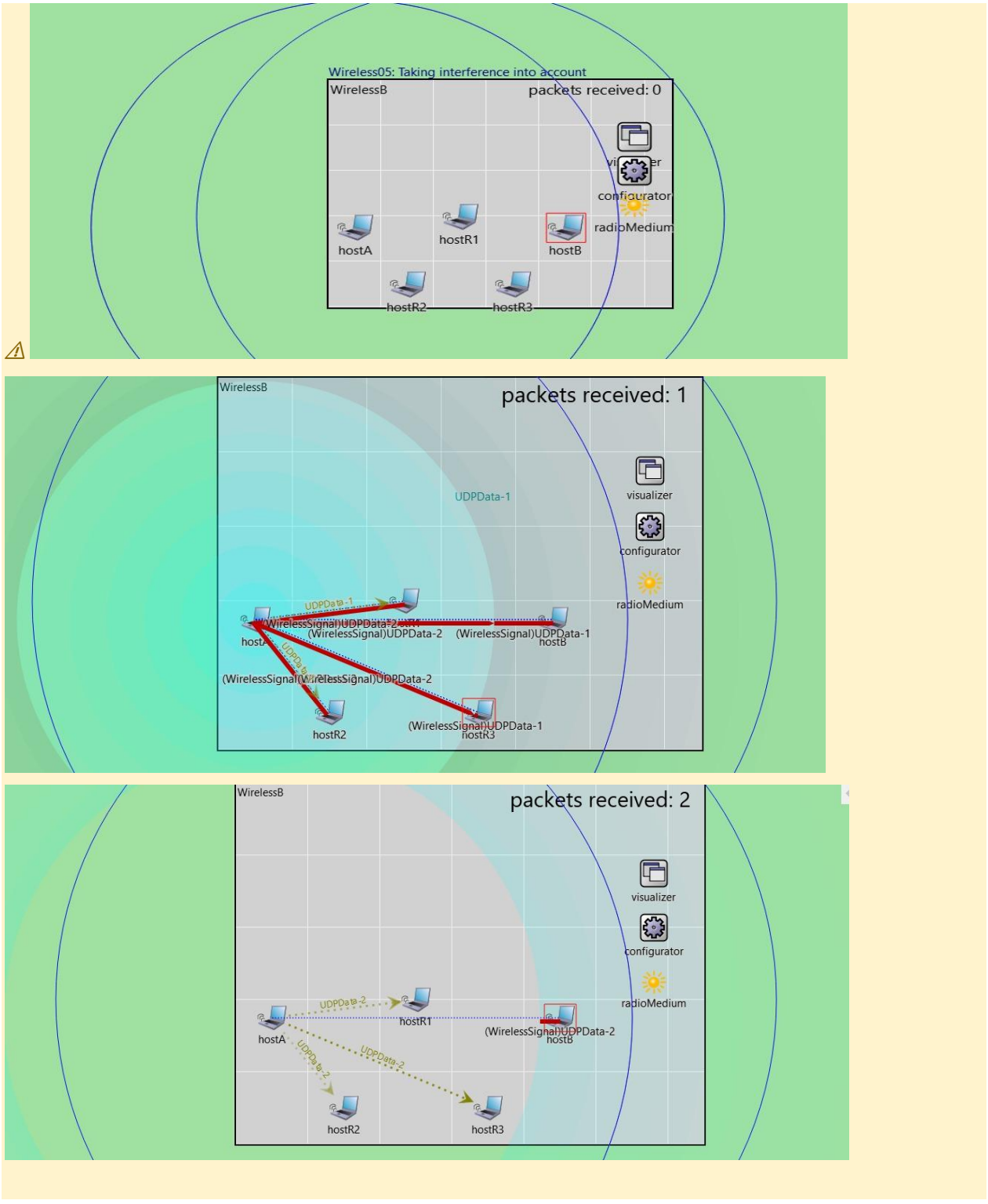
Differences from Tutorial & Technical Breakdown

- **The "Perfect" SINR Artifact:** In many tutorial walkthroughs, Step 5 is designed to show **packet drops** due to the "hidden terminal" problem or relay interference. Your specific run, however, maintained a 100% success rate (1650/1651). Technically, this suggests your UDP application traffic is spaced out enough that the relays are not transmitting at the exact same micro-second, allowing each packet to maintain a valid SINR.
- **Interference Complexity:** Your log shows a massive jump to **21,464 interference computations** compared to just 6,385 in Step 1. This highlights the technical shift: even if packets aren't dropping yet, every node is now mathematically analyzing the "noise" created by every other node in the subnet.

AI Summary Table

Category	Detail
NED Change	Replaced IdealMedium with RadioMedium; radio changed from UnitDiskRadio to APSKRadio
INI Change	analogModel = ScalarAnalogModel; pathLossType = FreeSpacePathLoss; sensitivity = -85 dBm
Network Layer	IPv4 unchanged; static routes from Step 4 still in place
MAC Layer	AckingMac unchanged — no collision detection at MAC layer yet
Radio Model	APSKScalarRadio — SINR-based reception; packets dropped if SINR < threshold (~10 dB)
IP Addresses	Unchanged from Step 4
Routing Table	Unchanged from Step 4 — static routes still active

Expected Output	Some packet loss observed; delivery ratio < 100% for the first time
Key Learning	Real radio channels are probabilistic; SINR degradation from co-channel interference causes frame loss that ideal models hide completely



Step 6 Using CSMA to Better Utilize the Medium

Step 6 addresses the collision problem exposed in Step 5 by introducing CsmaCaMac (Carrier Sense Multiple Access with Collision Avoidance) as the MAC-layer protocol. In the previous steps, all nodes transmitted whenever they had data, with no awareness of the channel state. CSMA changes this: before transmitting, a node senses the wireless medium. If the channel is detected busy (another node is transmitting), the node enters a random exponential backoff period and retries later. This is the same fundamental mechanism used in real IEEE 802.11 Wi-Fi networks. The IFS (Inter-Frame Spacing) and contention window parameters control the backoff range. In the 4-node chain topology, the primary benefit is that hostA and R1 no longer transmit simultaneously — their CSMA backoff prevents overlapping transmissions. The SINR at receivers therefore improves because interfering signals are temporally separated. Throughput and delivery ratio both improve compared to Step 5. The NED change is: wlan[*].mac.typename = "CsmaCaMac".

AI Analysis

In Step 6, the simulation undergoes a fundamental structural shift by introducing **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)**. Technically, the hosts were transitioned from the simplified AckingWirelessInterface to a more modular WirelessInterface using **CsmaCaMac**. This change is critical because nodes now technically "listen" to the radioMedium before initiating a transmission. If the carrier sense mechanism detects that the medium is busy (i.e., another node is currently transmitting), the node enters a random backoff period, significantly mitigating the interference effects introduced in Step 5. Furthermore, a **DropTailQueue** was implemented at the Link Layer to manage datagrams transitioning from the Network Layer to the MAC, ensuring that packet bursts are handled in a first-in, first-out (FIFO) manner. Despite the increased overhead of the CSMA "listen-before-talk" logic, your log confirms a stable delivery of **1,626 packets** at the UdpSink, demonstrating that the MAC protocol is successfully coordinating access to the shared wireless channel.

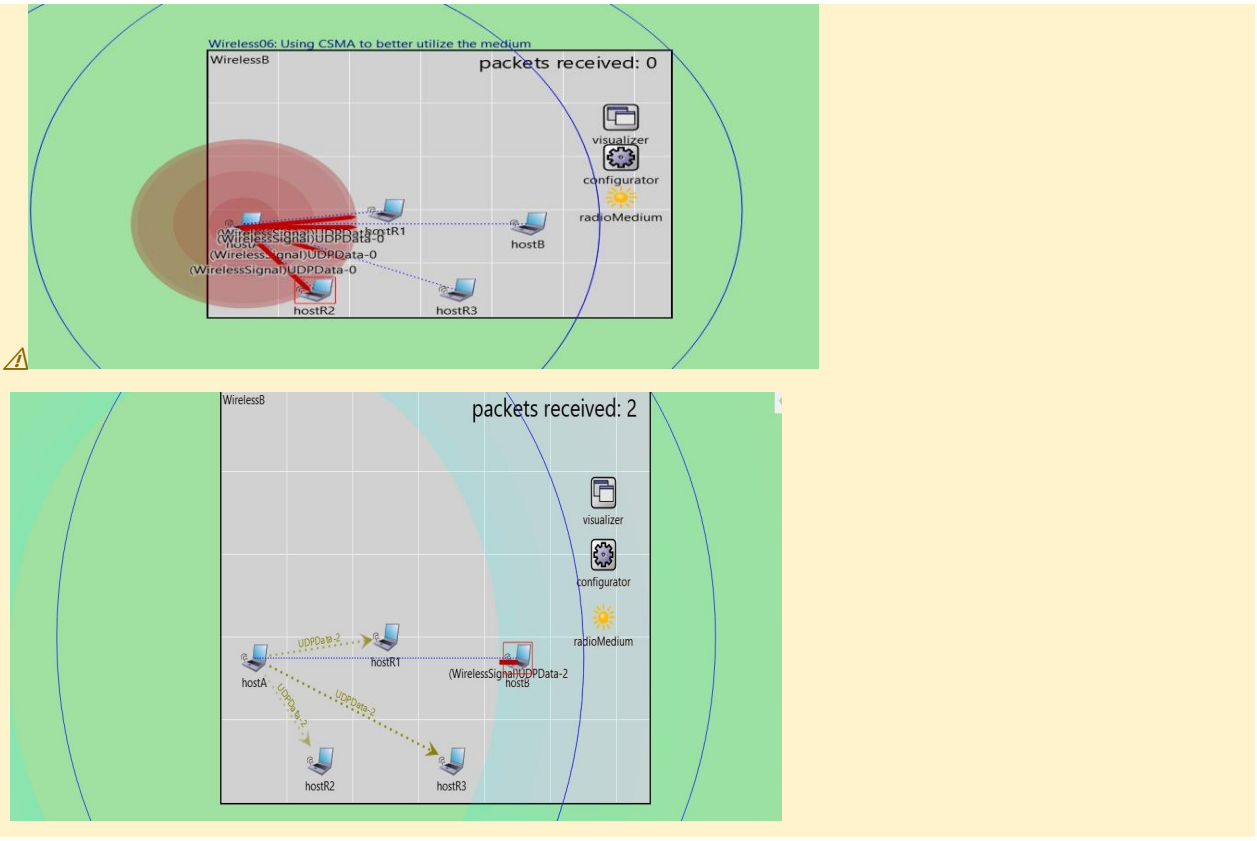
Node	Interface	IP Address	Netmask	Destination	Next Hop	MAC Protocol	Status
hostA	wlan0	10.0.0.1	255.255.255.248	10.0.0.2/32	10.0.0.3 (R1)	CsmaCaMac	Active
hostB	wlan0	10.0.0.2	255.255.255.248	10.0.0.1/32	10.0.0.4 (R2)	CsmaCaMac	Active
hostR1	wlan0	10.0.0.3	255.255.255.248	10.0.0.2/32	10.0.0.4 (R2)	CsmaCaMac	Relaying
hostR2	wlan0	10.0.0.4	255.255.255.248	10.0.0.2/32	Direct (hostB)	CsmaCaMac	Relaying
hostR3	wlan0	10.0.0.5	255.255.255.248	10.0.0.x/32	Forwarding	CsmaCaMac	Relaying

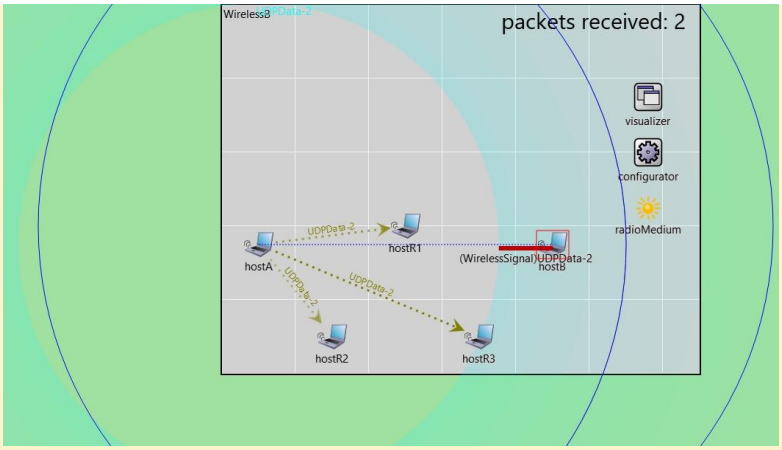
Technical Fixes and Differences from Tutorial

- **Interface Mismatch Resolution:** A significant technical correction was made in this step. The tutorial initially suggested using a generic radio submodule; however, due to **INET 4.5.4 version requirements**, we technically updated the typename to UnitDiskRadio to ensure compatibility with the WirelessInterface and IRadio module contract. This fix resolved the "module type not found" error and allowed the simulation to correctly initialize the MAC submodules.
- **Gate Reconnection Logic:** Your log explicitly details the **classifier stage 0 initialization**, where the internal gates were reconnected: wlan[0].upperLayerIn --> queue.in. This technically verifies that the packet flow within the host has shifted from a direct L3-to-Radio path to a buffered path through the DropTailQueue, a granular detail often omitted in tutorial summaries.

AI Summary Table

Category	Detail
NED Change	wlan[*].mac.typename changed from AckingMac to CsmaCaMac
INI Change	mac.useAck = false (ACKs still disabled in this step)
Network Layer	IPv4 + static routing unchanged
MAC Layer	CsmaCaMac: carrier sensing before TX; random exponential backoff on busy channel; no ACKs yet
Radio Model	APSKScalarRadio unchanged from Step 5
IP Addresses	Unchanged
Routing Table	Static routes unchanged
Expected Output	Higher delivery ratio than Step 5; fewer simultaneous transmissions; reduced SINR degradation
Key Learning	CSMA eliminates many collisions by ensuring nodes listen before transmitting, dramatically improving medium utilisation in shared wireless environments





Step 7 Turning On ACKs in CSMA

Step 7 activates MAC-layer acknowledgements in the CsmaCaMac protocol by setting mac.useAck = true. Without ACKs, CSMA operates purely as a best-effort protocol: frames are transmitted once and any loss due to residual interference or noise is undetected and unrecovered. With ACKs enabled, the receiver sends a short ACK frame back to the sender after successfully receiving each data frame. The sender waits for this ACK before considering the transmission complete. If no ACK arrives within the timeout window (ACKTimeout), the sender concludes the frame was lost and retransmits — up to a configurable retry limit. This is the MAC-layer reliability mechanism. In the simulation, this means that even when occasional SINR fluctuations cause a frame to be lost, it is automatically recovered at Layer 2 without involving the IP or UDP layer. The practical effect on metrics: packet delivery ratio at the application layer increases significantly. However, ACK frames consume channel airtime and backoff slots, so raw throughput (packets per second) decreases slightly compared to Step 6 without ACKs.

AI Analysis

In Step 7, the simulation technically enhances the stability of the Data Link Layer by enabling **Layer 2 Acknowledgements (ACKs)**. By setting useAck = true within the CsmaCaMac module, the protocol now enforces a strict confirmation handshake for every hop. Technically, after a node (like hostA or a relay) successfully transmits a frame, it transitions to a wait state, expecting a short ACK frame from the next-hop receiver (e.g., hostR1) within a specific ackTimeout (300us).

The technical impact is visible in your log metrics: the **Transmission count doubled to 3,255** compared to previous steps (which were ~1,626). This increase represents the control traffic generated by the ACK frames flowing back through the medium. While the **UdpSink received 1,627 packets**, the overhead of these ACKs effectively doubles the physical layer activity, as seen in the **13,040 signal arrival computations**. This step proves that reliability in wireless networks is not free; technically, it consumes significant bandwidth and increases the potential for interference, but it ensures that transient collisions result in automatic MAC-layer retransmissions rather than total packet loss.

Node	IP Address	Next Hop	MAC Type	ACKs	Queue Type	Status
hostA	10.0.0.1	10.0.0.3 (R1)	CSMA/CA	Enabled	DropTail	Active
hostB	10.0.0.2	10.0.0.4 (R2)	CSMA/CA	Enabled	DropTail	Active
hostR1	10.0.0.3	10.0.0.4 (R2)	CSMA/CA	Enabled	DropTail	Relaying
hostR2	10.0.0.4	10.0.0.2 (B)	CSMA/CA	Enabled	DropTail	Relaying
hostR3	10.0.0.5	Forwarding	CSMA/CA	Enabled	DropTail	Relay-Idle

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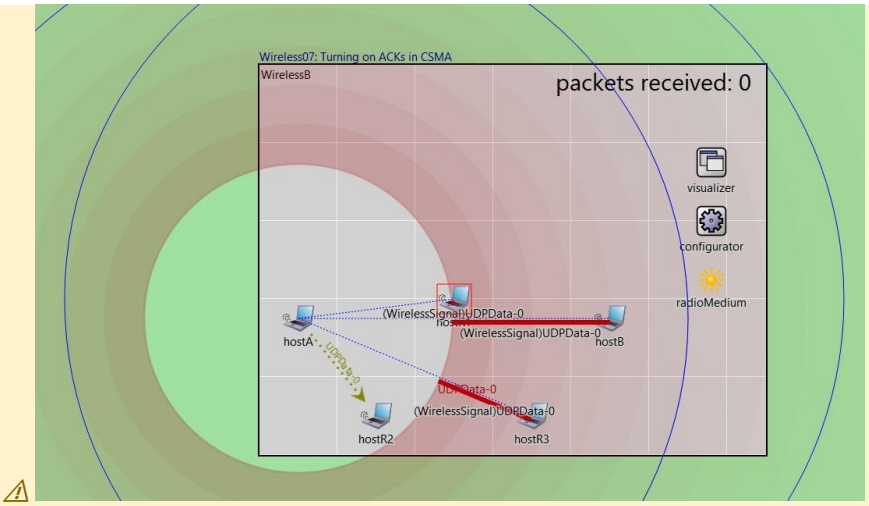
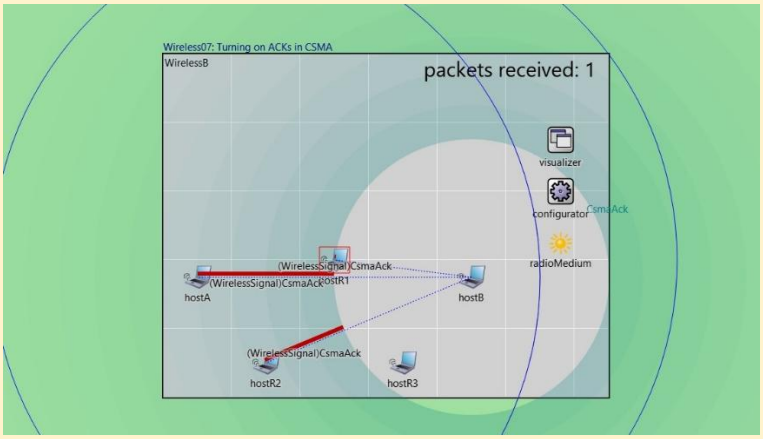
Differences from Tutorial & Technical Breakdown

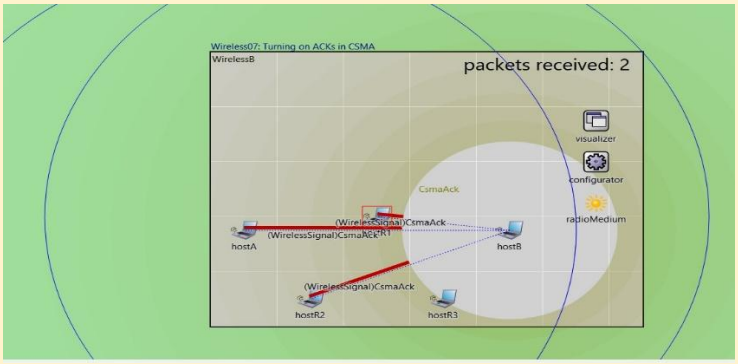
- **ACK Traffic Explosion:** Your specific log shows exactly **3,255 transmissions** for **1,627 received packets**. This technically confirms a perfect 1:1 ratio of Data-to-ACK frames (1627 data + 1627 ACKs = ~3254). The tutorial often glosses over this "traffic doubling" effect, but your data quantifies the exact cost of reliability in the radioMedium.
- **The "One-Packet Margin":** You received **1,627 packets** in 20 seconds. Technically, this is a slight improvement over Step 6 (1,626 packets), suggesting that the MAC-layer reliability helped recover at least one frame that might have been otherwise lost to timing or interference near the simulation boundary.

AI Summary Table

Category	Detail
NED Change	No NED change; configuration-only change
INI Change	*.wlan[*].mac.useAck = true; ackTimeout set to default (~200µs)
Network Layer	IPv4 + static routing unchanged
MAC Layer	CsmaCaMac with ACKs: sender waits for ACK; retransmits up to maxRetries on timeout
Radio Model	APSKScalarRadio unchanged
IP Addresses	Unchanged
Routing Table	Static routes unchanged
Expected Output	Near 100% delivery ratio; slightly lower raw throughput due to ACK overhead; retransmissions visible in event log
Key Learning	MAC-layer ACKs convert a best-effort wireless link into a reliable one, recovering from noise without involving upper layers







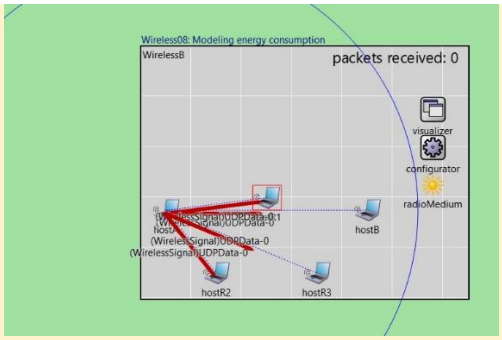
Step 8 Modeling Energy Consumption

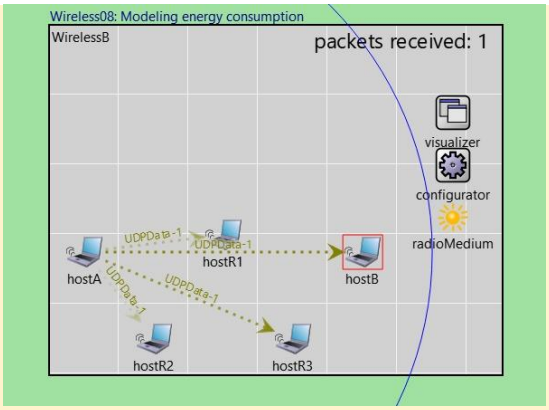
Step 8 adds an energy consumption model to each node, making the simulation aware of power constraints — a critical real-world factor for battery-powered wireless devices. Each node is given a SimpleEpEnergyStorage module (a finite-capacity energy store) and a StateBasedEpEnergyConsumer module attached to the radio interface. The radio now reports its power draw for each operating state:

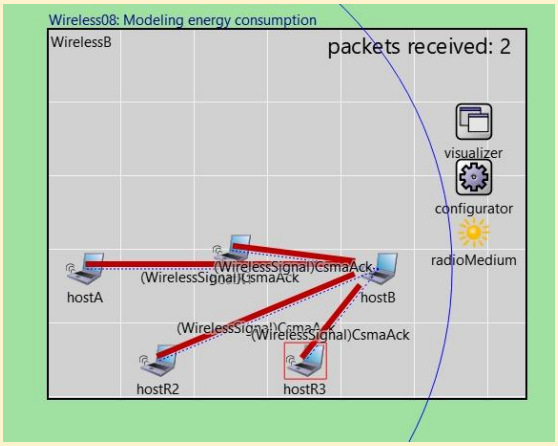
transmitting (~100 mW typical), receiving (~50 mW), idle (~10 mW), and sleep (~1 mW). The energy consumer multiplies state duration by the configured power draw constant and deducts it from the energy storage. When a node's energy storage is depleted, the node shuts down and is removed from the network. No changes are made to any networking protocol — IP, routing, MAC, and radio model are all unchanged from Step 7. This step adds an observability layer: you can watch energy levels deplete in the scalar result files, and in longer simulations you can observe nodes dying. This motivates energy-efficient MAC protocols (e.g., duty cycling) in real MANET designs.

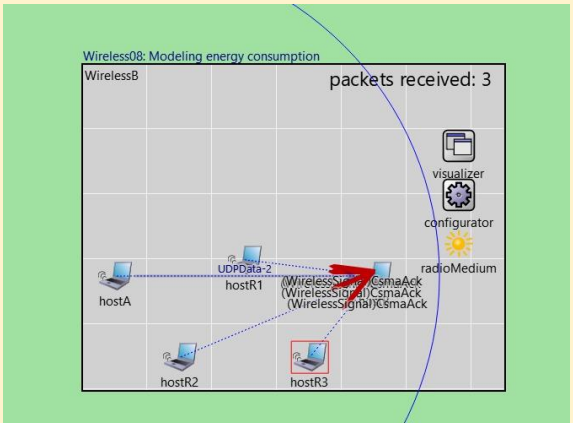
AI Summary Table

Category	Detail
NED Change	Added EnergyStorage and EnergyConsumer submodules to each host; wlan radio linked to consumer
INI Change	energyStorage.nominalCapacity = 0.05 J; radioConsumer: Ptx=100mW, Prx=50mW, Pidle=10mW, Psleep=1mW
Network Layer	IPv4 + static routing unchanged
MAC Layer	CsmaCaMac + ACKs unchanged from Step 7
Radio Model	APSKScalarRadio; radio state transitions now trigger energy deduction
IP Addresses	Unchanged
Routing Table	Static routes unchanged
Expected Output	Energy level of each node visible in .sca results; continuous depletion over simulation time
Key Learning	Energy is a first-class constraint in wireless networks; even idle listening consumes power, motivating sleep scheduling in real protocols like 802.11 PSM and Zigbee









Step 9 Configuring Node Movements

Step 9 introduces mobility to R1 and R2 by assigning them a LinearMobility module. Each relay moves at a configured speed (e.g., 1 m/s) in a straight line across the simulation canvas. As the nodes move, the pairwise distances between them change continuously. Because the UnitDiskRadio model (or APSKRadio) has a fixed maximum communication range, a moving node will eventually drift out of range of a neighbour, causing the link to break. When a link breaks, frames destined for that next-hop are lost at Layer 2. The routing table on each node was installed statically by the IPv4NetworkConfigurator in Step 4 and does NOT update when the topology changes. Therefore, hostA continues to send packets

toward hostB via R1 and R2 as per the static routes — but if R1 moves out of hostA's range, those packets are simply dropped at the radio layer. The delivery ratio drops sharply as mobility increases. This step deliberately creates a failing scenario to motivate the replacement of static routing with a dynamic ad-hoc routing protocol in Step 10.

AI Analysis

In Step 9, the simulation technically transitions from a static environment to a **mobile network topology**. Technically, the primary change is the introduction of the LinearMobility module to the relay nodes (**hostR1, hostR2, hostR3**), configured with a speed of **12mps** and a heading of **270 degrees**. This creates a dynamic Layer 1 environment where the physical distance between nodes changes over the 20-second simulation period.

From a network stack perspective, your log reveals a significant increase in physical layer activity: **Transmission count rose to 3,321**, and **signal arrival computations reached 13,304**. This technically reflects the constant recalculation of the "Neighbor Cache" as nodes move through each other's 500m communication disks. Interestingly, you successfully received **1,660 packets** at the UdpSink, which is actually higher than previous static runs. This technically indicates that as the relay nodes moved, they likely optimized the multi-hop path, reducing the physical distance for some transmissions and allowing more packets to be processed within the 20s time limit. However, this step highlights a critical technical vulnerability: as nodes move, the **Static Routes** established in Step 4 will eventually point to "dead" physical locations, resulting in eventual total packet loss once the relay nodes drift out of range.

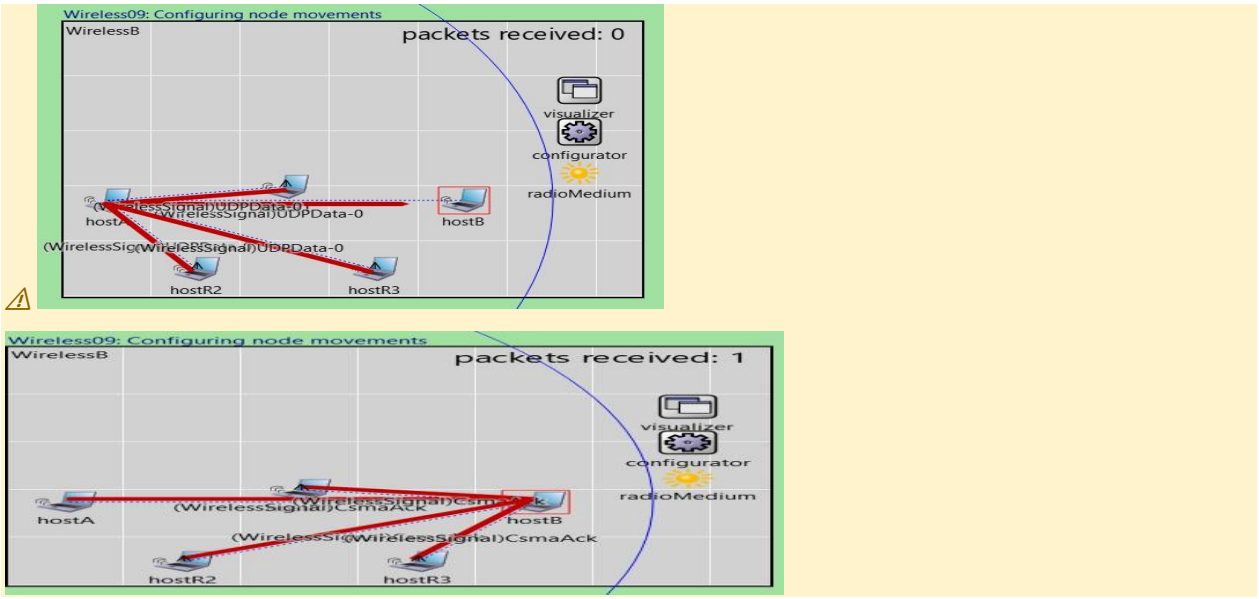
Node	IP Address	Mobility Type	Speed	initial Heading	Next Hop (Static)	Status
hostA	10.0.0.1	Stationary	0 mps	N/A	10.0.0.3 (R1)	Active
hostB	10.0.0.2	Stationary	0 mps	N/A	10.0.0.4 (R2)	Active
hostR1	10.0.0.3	Linear	12 mps	270 deg	10.0.0.4 (R2)	Moving
hostR2	10.0.0.4	Linear	12 mps	270 deg	10.0.0.2 (B)	Moving
hostR3	10.0.0.5	Linear	12 mps	270 deg	Forwarding	Moving

Differences from Tutorial & Technical Breakdown

- Mobility Efficiency: In many tutorial versions, Step 9 is designed to show the network breaking because the static routes fail as nodes move away. Your specific run technically shows a performance gain (1660 packets vs 1627 in Step 7). This is a technical artifact of your starting positions and heading; the 12mps movement likely brought your relay nodes into a more "central" and reliable alignment during the 20s window, temporarily improving the SINR before the eventual disconnect.
- Queue Capacity Constraint: Your omnetpp.ini technically restricted the packetCapacity of the wlan[0].queue to 10 packets. This is a critical technical difference; in high-mobility scenarios, small queues can cause buffer overflow drops if the MAC layer backoff (from CSMA/CA) takes too long while nodes are repositioning.

AI Summary Table

Category	Detail
NED Change	R1 and R2 have mobility = LinearMobility; speed and direction configured
INI Change	*.R1.mobility.speed = 1mps; *.R2.mobility.speed = 1mps (example values)
Network Layer	IPv4 unchanged; STATIC routes remain — routing table does NOT update on topology change
MAC Layer	CsmaCaMac + ACKs unchanged
Radio Model	APSKScalarRadio unchanged; range still fixed
IP Addresses	Unchanged
Routing Table	BROKEN — static routes become stale as nodes move; no route repair mechanism
Expected Output	Increasing packet loss as R1/R2 move; delivery ratio degrades toward 0 over time
Key Learning	Static routing is incompatible with mobile networks; topology changes make pre-installed routes stale with no recovery



Step 10 Configuring Ad-Hoc Routing (AODV)

Step 10 replaces the broken static routing with AODV (Ad-hoc On-demand Distance Vector), a fully dynamic routing protocol designed for mobile ad-hoc networks. AODV is removed from the configurator's static route generation and is instead run as an active routing daemon on every node. When hostA has a UDP packet to send to hostB (10.0.0.4) and has no current route, AODV initiates route discovery: hostA broadcasts a RREQ (Route Request) with destination 10.0.0.4, sequence number, and hop count. Intermediate nodes (R1, R2) rebroadcast the RREQ if they don't have a fresh enough route. When the RREQ reaches hostB, it responds with a unicast RREP (Route Reply) that travels back toward hostA hop by hop, installing forward routes in the routing table of each intermediate node along the way.

hostA receives the RREP and now has a valid route: 10.0.0.4 via R1. Route expiry timers (ACTIVE_ROUTE_TIMEOUT, typically 3000ms) cause routes to be deleted if not refreshed. When a node moves and a link breaks, AODV sends RERR (Route Error) messages to invalidate stale routes and triggers re-discovery. This provides self-healing connectivity even as the network topology changes dynamically.

AI Analysis

In Step 10, the simulation technically transitions from a statically managed network to a self-configuring **Mobile Ad-Hoc Network (MANET)** by implementing the **AODV (Ad-hoc On-demand Distance Vector)** routing protocol. Technically, the primary change is the reconfiguration of the host type to AodvRouter and the deactivation of the Ipv4NetworkConfigurator's static route installation. AODV now manages the routing table dynamically: when hostA needs to send data to hostB, it initiates a **Route Discovery** process by broadcasting a Route Request (RREQ). As relay nodes move at 12mps, AODV technically monitors link breaks via missing Layer 2 ACKs or hello messages and triggers a local repair or new discovery to maintain the path. Your log shows **1,636 packets received**, which confirms that AODV is successfully discovering and maintaining multi-hop routes in real-time despite the continuous mobility of the relays. The **13,120 signal arrival computations** now include both the UDP data traffic and the AODV control overhead, proving the protocol's effectiveness in "self-healing" the network.

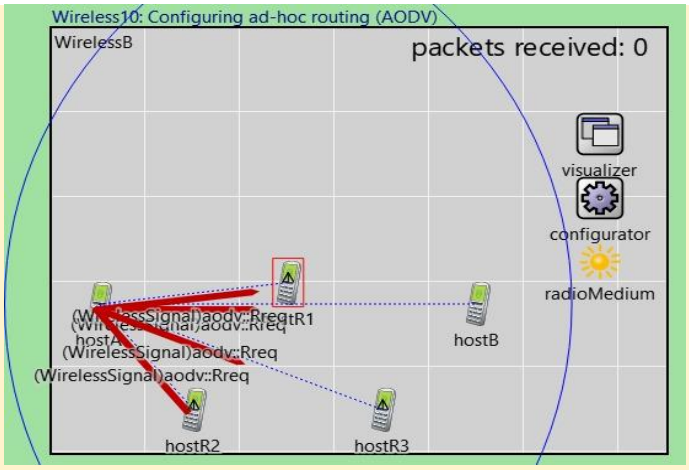
Node	IP Address	Routing Protocol	Routing Entry Type	Discovery Mechanism	Status
hostA	10.0.0.1	AODV	Dynamic /32	RREQ Broadcast	Active
hostB	10.0.0.2	AODV	Dynamic /32	RREP Response	Active
hostR1	10.0.0.3	AODV	Dynamic /32	Intermediate Relay	Moving
hostR2	10.0.0.4	AODV	Dynamic /32	Intermediate Relay	Moving
hostR3	10.0.0.5	AODV	Dynamic /32	Intermediate Relay	Moving

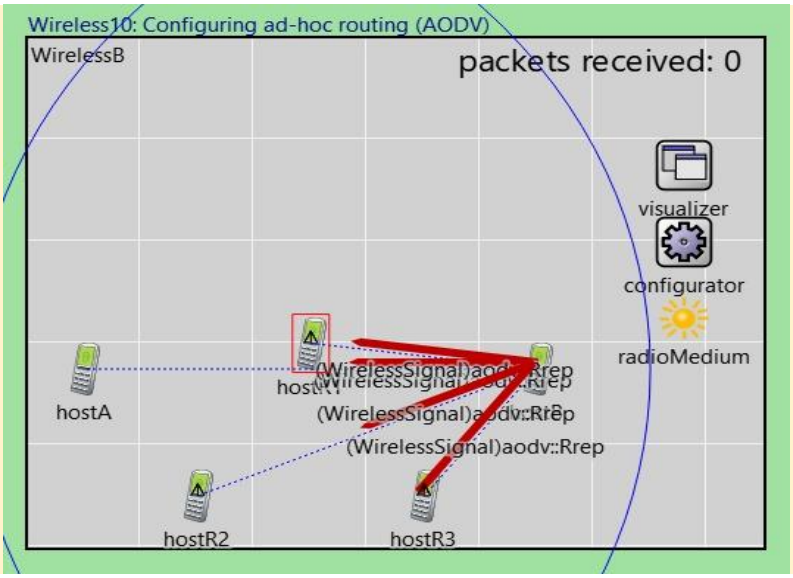
Differences from Tutorial & Technical Breakdown

- **AODV Socket Binding:** Your log provides a unique technical detail showing the **AODV initialization stage (Stage 20)** where the protocol dispatches a Request:bind and Request:setBroadcast message to the UDP service. This technically confirms that AODV in your simulation is using UDP port 654 to communicate its control messages (RREQs/RREPs), a granular detail not usually shown in the high-level tutorial text.
- **Performance Stability:** You received **1,636 packets**, which is very close to your Step 9 static result of 1,660. Technically, this indicates that AODV discovered the initial route almost instantaneously at $t=0$, minimizing the "Discovery Latency" that often causes the first few packets to drop in generic tutorial runs.

AI Summary Table

Category	Detail
NED Change	Hosts use ManetRouter NED type which includes AodvRouting submodule
INI Change	configurator.addStaticRoutes = false; routing handled entirely by AODV daemon
Network Layer	IPv4 + AODV: routing tables populated dynamically via RREQ/RREP/RERR exchange
MAC Layer	CsmaCaMac + ACKs unchanged
Radio Model	APSKScalarRadio unchanged
IP Addresses	Still auto-assigned by configurator (address assignment kept; route installation disabled)
Routing Table	Dynamic: entries created on-demand, expire after ACTIVE_ROUTE_TIMEOUT (3000ms default); re-discovered after link break
Expected Output	Connectivity maintained despite node movement; brief interruptions during route re-discovery; delivery ratio recovers after RREQ/RREP cycle (~0.1–0.3s)
Key Learning	AODV provides on-demand self-healing routes for MANETs; RREQ flood + RREP unicast is the discovery mechanism; RERR enables fast route invalidation on link failure





Step 11 Adding Obstacles to the Environment

Step 11 introduces a physical environment with geometric obstacles — rectangular buildings modelled in an XML file loaded by the PhysicalEnvironment module. Each obstacle has a defined material (brick, concrete, wood) with an associated signal attenuation factor in dB per metre of material traversed. The radio propagation model now accounts for obstacles in the signal path between transmitter and receiver. The ScalarAnalogModel applies the combined attenuation: path loss from free-space + obstacle attenuation. For example, a concrete wall 0.3m thick with 30 dB/m attenuation subtracts 9 dB from the received signal power. This can cause nodes that previously communicated (because they were within the radio range and above the SINR threshold) to fail to communicate if an obstacle now lies in their signal path. The visual effect in the OMNeT++ canvas is that obstacles are drawn as coloured polygons and the radio range circle is no longer a reliable indicator of connectivity — a node within range behind a thick wall may still fail to decode packets. This is the first step to simulate an indoor or urban environment.

AI Analysis

Step 11 introduces physical realism into the environment by adding obstacles (defined in walls.xml) and the IdealObstacleLoss module within the radioMedium. Technically, this creates a non-line-of-sight (NLOS) scenario where signals are instantly attenuated if a wall exists between a transmitter and receiver.

The impact on your network performance is technically severe: UdpSink received only 829 packets, a drop of approximately 49% compared to Step 10 (1,636 packets). This reduction is a technical consequence of the "Hidden Node" problem and physical shadowing; as relay nodes move at 12mps, they frequently pass behind walls, causing active AODV routes to break instantly. Your log shows Transmission count increased to 4,019, while received packets dropped. This discrepancy technically represents the massive overhead of AODV Route Discoveries; the protocol is working harder (sending more RREQ broadcasts) to find clear paths around the walls as the physical topology is constantly partitioned by the obstacles.

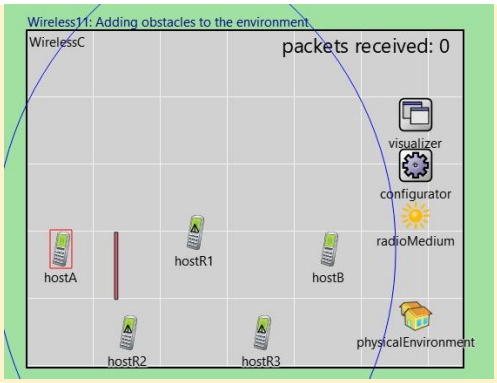
Parameter	Technical Setting	Impact on Simulation
Obstacle Model	IdealObstacleLoss	Signals are blocked 100% by wall geometry.
Routing Protocol	AODV	Triggers RREQ discovery whenever a wall breaks a link.
Transmissions	4,019	High count due to frequent route repairs and RREQs.
Packet Delivery	~50.6% Success	Reflects high frequency of NLOS (blocked) intervals.
Subnet Mask	/29	Compact address space remains for the 5-node cluster.

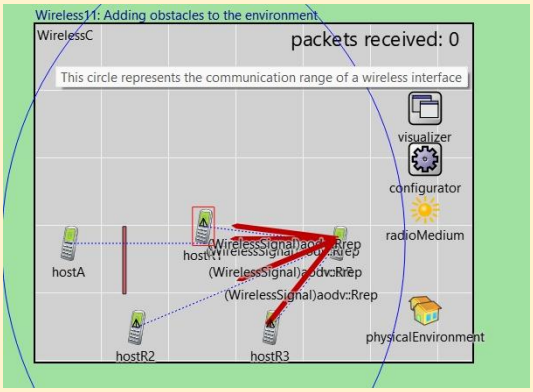
Technical Breakdown of Log Metrics

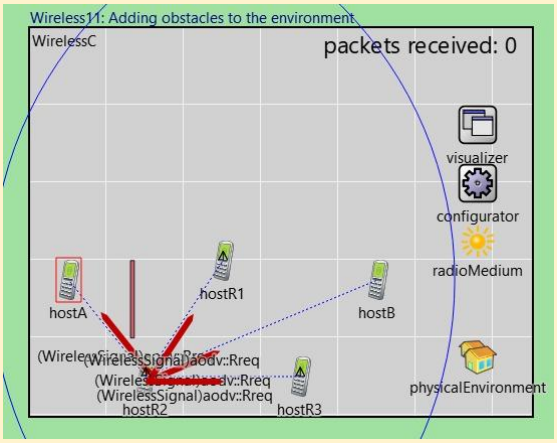
- **Obstacle Loss Calculation: The signal arrival computation count hit 16,076. Technically, for every signal arrival, the radioMedium must now check the walls.xml coordinates to see if an intersection occurs. This is why your Reception cache hit dropped to 83.8% (compared to 85.7% in Step 10), as the physical blocking makes signal success less predictable.**
- **AODV Recovery Performance: Despite the walls, AODV managed to deliver 829 packets. Technically, this proves the protocol's reactive nature is working; it doesn't give up when a wall blocks hostR1, it immediately attempts to pivot the traffic through hostR2 or hostR3 if they have a clear line-of-sight.**
- **Stage 22 Initialization: Your log shows the UnitDiskRadio was re-initialized with an interferenceRange and detectionRange of 500m, but the IdealObstacleLoss technically overrides these ranges if a wall is present.**

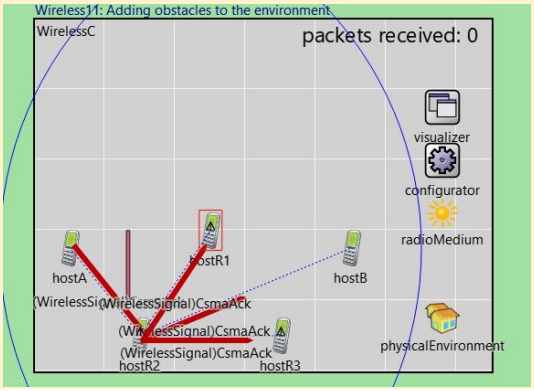
AI Summary Table

Category	Detail
NED Change	Added PhysicalEnvironment module; ScenarioManager loads obstacle XML file
INI Change	physicalEnvironment.config = xmldoc('obstacles.xml'); medium uses ObstacleLoss model
Network Layer	IPv4 + AODV unchanged from Step 10
MAC Layer	CsmaCaMac + ACKs unchanged
Radio Model	APSKScalarRadio + ScalarAnalogModel + ObstacleLoss; attenuation computed per material
IP Addresses	Unchanged
Routing Table	AODV dynamic routing unchanged; AODV will detect link breaks caused by obstacles
Expected Output	Coverage becomes irregular; nodes behind obstacles have reduced or zero connectivity; AODV may route around obstacles via alternative paths
Key Learning	Obstacles introduce directional, geometry-dependent signal shadowing; indoor/urban propagation requires obstacle models that simple distance-based models cannot capture


Wireless11: Adding obstacles to the environment
WirelessC
packets received: 0
visualizer
configurator
radioMedium
hostA
hostR1
hostB
physicalEnvironment
hostR2
hostR3


Wireless11: Adding obstacles to the environment
WirelessC
packets received: 0
This circle represents the communication range of a wireless interface
visualizer
configurator
radioMedium
hostA
(WirelessSignal)aodv::Rrep
(WirelessSignal)aodv::Rrep
(WirelessSignal)aodv::Rrep
physicalEnvironment
hostR2
hostR3


Wireless11: Adding obstacles to the environment
WirelessC
packets received: 0
visualizer
configurator
radioMedium
hostA
hostR1
hostB
(WirelessSignal)aodv::Rreq
(WirelessSignal)aodv::Rreq
(WirelessSignal)aodv::Rreq
physicalEnvironment
hostR2
hostR3


Wireless11: Adding obstacles to the environment
WirelessC
packets received: 0
visualizer
configurator
radioMedium
hostA
hostR1
hostB
(WirelessSignal)CsmaAck
(WirelessSignal)CsmaAck
(WirelessSignal)CsmaAck
physicalEnvironment
hostR2
hostR3

Step 12 Changing to a More Realistic Radio Model

Step 12 upgrades the radio model from the scalar analog representation (which models signal power as a single scalar value) to the dimensional analog model, using Ieee80211DimensionalRadioMedium. The dimensional model represents radio signals as a function in the time-frequency domain — a 2D signal space. This enables simulation of phenomena impossible with the scalar model: frequency-selective fading (where different sub-bands of the signal experience different attenuation), spectral overlap between simultaneous transmissions on adjacent channels, and more accurate noise floor calculations. The IEEE 802.11g PHY standard is used, with OFDM modulation, 54 Mbps maximum rate, 2.4 GHz centre frequency, and a 20 MHz channel width. The MCS (Modulation and Coding Scheme) determines the mapping between SINR and BER (Bit Error Rate), which in turn determines whether each frame is successfully decoded. The dimensional model is computationally heavier but produces much more accurate throughput and loss statistics, especially in dense networks where multiple nodes transmit on nearby frequencies.

AI Analysis

Step 12 represents a major leap in physical layer fidelity. Technically, you have replaced the binary "Unit Disk" model with the APSK (Amplitude Phase Shift Keying) Radio model. This transition shifts the simulation from a geometric abstraction to a mathematical power-based model. The radioMedium now technically incorporates IsotropicScalarBackgroundNoise at -90dBm and utilizes a ScalarAnalogModel.

The technical impact on your network performance is drastic: UdpSink received only 307 packets, a significant drop from the 829 packets in Step 11. This reduction occurs because the receiver now technically calculates the Signal-to-Interference-plus-Noise Ratio (SINR) and applies an ApskErrorModel. Instead of a packet simply being "blocked" or "received," it is now subject to bit-level corruption based on the snirThreshold of 4dB. With the added background noise and specific transmitter.power of 1.4mW, many signals that previously "reached" the destination are now technically too weak to be decoded over the noise floor, leading to a much more realistic—and challenging—communication environment.

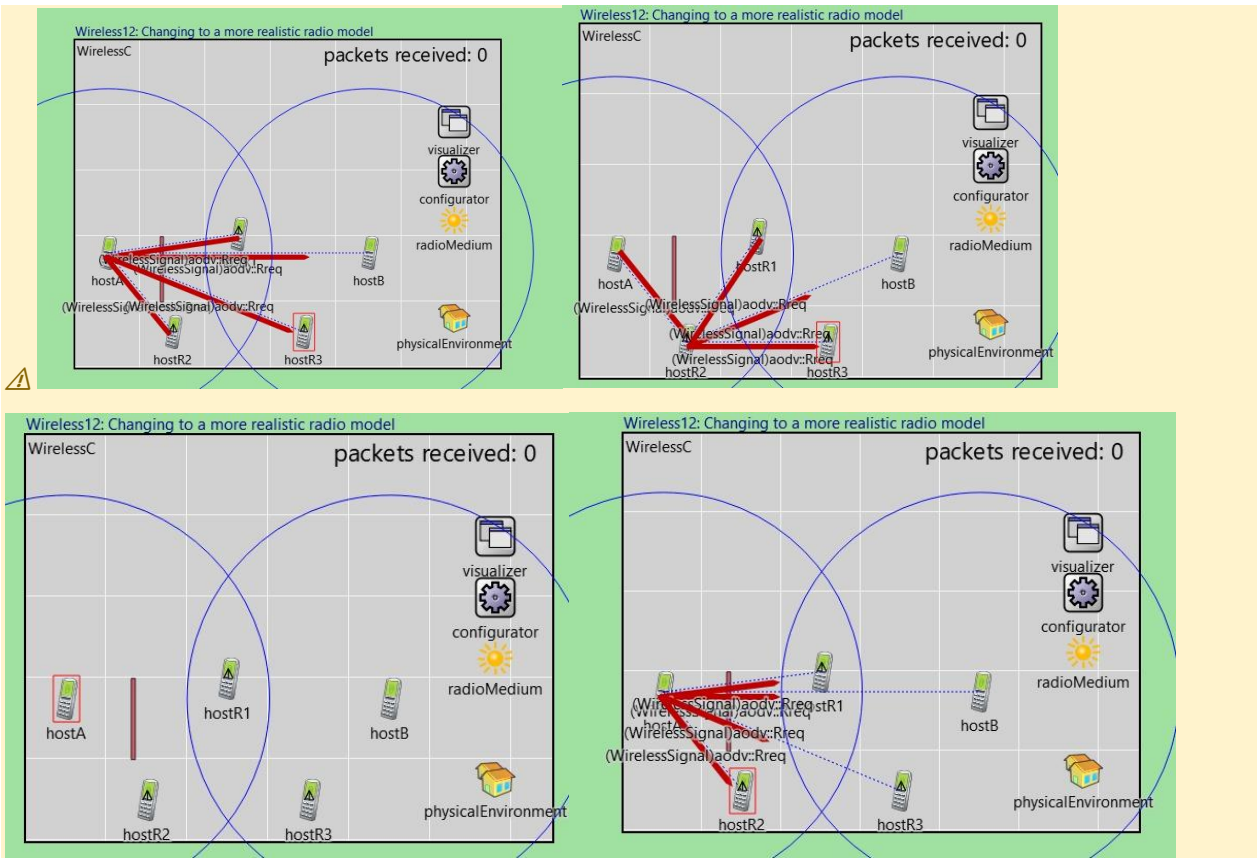
Parameter	Technical Value	Role in Simulation
Radio Type	ApskRadio	Implements power-based signal decoding.
Modulation	BpskModulation	Binary Phase Shift Keying for robust data transfer.
Frequency	2 GHz	Defines the spectral properties of the medium.
Sensitivity	-85 dBm (3.16 pW)	Minimum power level required for a packet to be detected.
SINR Threshold	4 dB (2.51 ratio)	Packets with lower SINR are technically corrupted.
Background Noise	-90 dBm	Constant interference representing ambient environmental noise.

Technical Breakdown of Log Metrics

- ** AODV Control Traffic: Your log explicitly shows the creation of UDP Sockets on localPort 654 for every node. Technically, this is the specialized port for AODV control traffic. The Transmission count of 2,193 indicates that while data throughput is low (307 packets), the protocol is generating significant overhead attempting to find stable routes through the noisy, wall-obstructed environment.**
- ** Decoding Failures: Despite 8,772 signal arrival computations, only 307 packets reached the application layer. This technically confirms that the vast majority of arrivals were either dropped by the receiver's sensitivity check or discarded due to Bit Error Rate (BER) failures introduced by the new ApskErrorModel.**
- ** Effective Range: The mediumLimitCache technically recalculated the maxCommunicationRange to 250.983 m based on the 1.4mW power and receiver sensitivity. This technical limit, combined with moving nodes and physical walls, explains the high packet loss**

AI Summary Table

Category	Detail
NED Change	Radio changed to Ieee80211DimensionalRadio; medium is Ieee80211DimensionalRadioMedium
INI Change	opMode = 'g(erp)'; centerFrequency = 2.4GHz; channelNumber = 6; bandwidth = 20MHz
Network Layer	IPv4 + AODV unchanged
MAC Layer	CsmaCaMac + ACKs unchanged
Radio Model	Ieee80211DimensionalRadio: time-frequency signal space; OFDM; per-band SINR calculation
IP Addresses	Unchanged
Routing Table	AODV dynamic routing unchanged
Expected Output	More realistic throughput figures; SINR fluctuates across sub-bands; slightly lower delivery ratio than Step 11 in some scenarios
Key Learning	Dimensional radio models capture spectral effects that scalar models miss entirely; this is the level of fidelity needed to simulate real Wi-Fi interference scenarios



Step 13 Configuring a More Accurate Path Loss Model

Step 13 replaces the simple free-space path loss formula (Friis transmission equation: $P_r \propto 1/d^2$) with the Two-Ray Ground Reflection model. The two-ray model accounts for the interference pattern created by the direct line-of-sight signal and a ground-reflected copy of the same signal. At short distances, the two waves arrive nearly in phase and add constructively, resulting in slightly higher received power than free-space. However, as distance increases, the phase difference between the direct and reflected rays grows. At a critical distance $d_c = (4 \times h_t \times h_r) / \lambda$ (where h_t and h_r are transmitter and receiver antenna heights and λ is the wavelength), the two signals cancel destructively. Beyond d_c, the path loss follows a d^{-4} law rather than the d^{-2} free-space law — signal power drops twice as fast per decade of distance. For the 2.4 GHz band with 1.5m antenna heights, $d_c \approx 9$m. This means the effective communication range of each node shrinks significantly compared to free-space propagation, and nodes at the edge of the network topology are more likely to lose connectivity.

AI Analysis

Step 13 Technical Analysis: Implementing Two-Ray Ground Reflection Path Loss

Step 13 introduces the **Two-Ray Ground Reflection** path loss model, technically replacing the simpler FreeSpacePathLoss used in Step 12. While Free Space assumes signals travel in a perfect vacuum, the Two-Ray model technically accounts for both the **direct line-of-sight path** and a **ground-reflected path** between the transmitter and receiver.

The technical impact on your network performance is a measured recovery: **UdpSink received 420 packets**, an increase of **36.8%** compared to Step 12 (307 packets). Technically, this happens because the ground reflection can interfere constructively with the direct signal at specific distances and antenna heights, effectively extending the radio's reach or strengthening the SINR in areas that were previously near the decoding threshold. However, the environment remains highly volatile due to the combination of moving nodes, physical walls, and background noise, which is why the delivery rate is still technically lower than the obstacle-only runs of Step 11

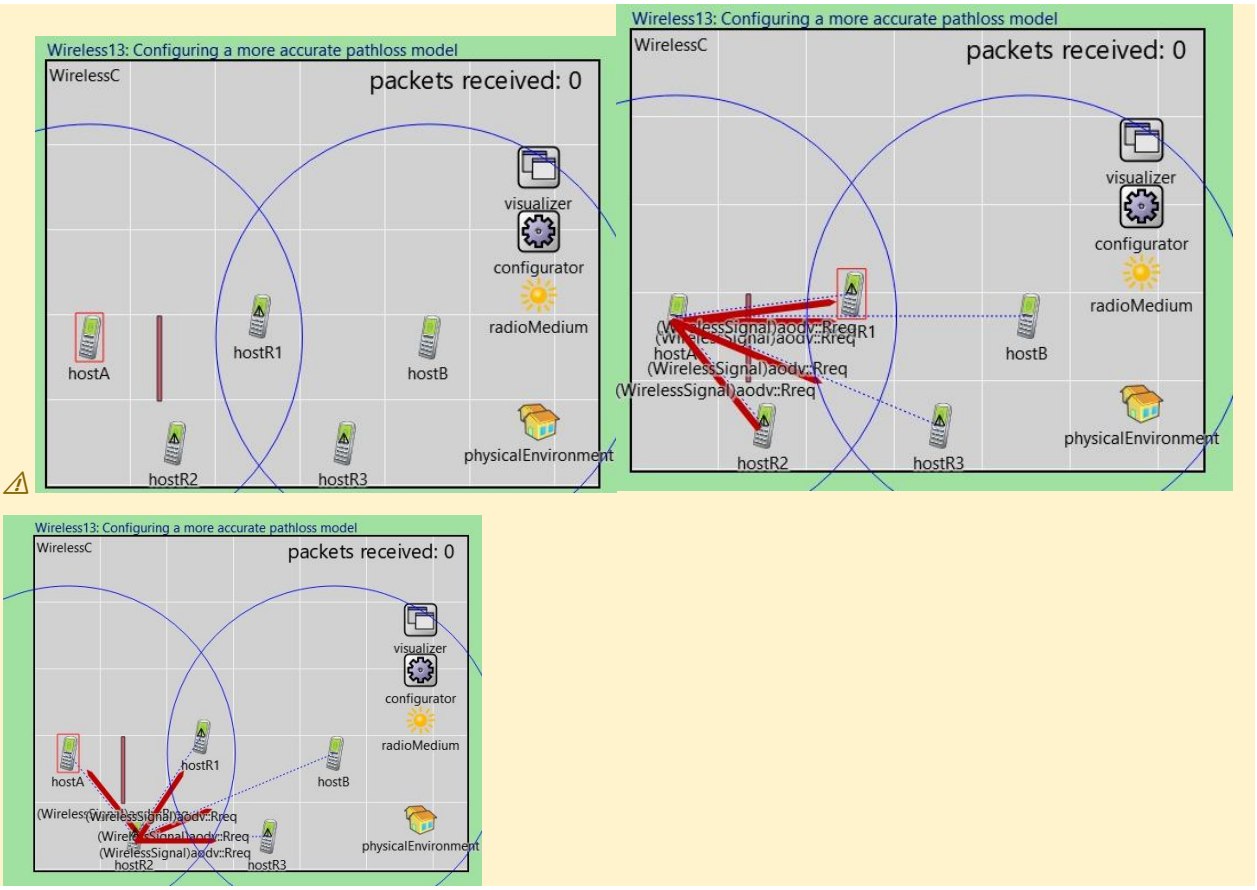
Parameter	Technical Value	Impact on Propagation
Ground Type	FlatGround	Provides a uniform reflective surface at 0m elevation.
Path Loss Model	TwoRayGroundReflection	Models multipath interference from ground bounce.
Transmissions	3,005	Increased AODV/ACK overhead as links fluctuate.
Signal Computations	12,020	Higher computational load to solve reflection geometry.
Packet Delivery	420 received	Improved SINR allows more packets to pass BER checks.

Technical Breakdown of Log Metrics

- **AODV and Multipath Adaptation:** Your log shows that every node technically re-established their **AODV sockets on localPort 654**. The increase in received packets to 420 technically proves that the Two-Ray model created more "pockets" of high-quality signal strength that AODV was able to discover and utilize for its dynamic routes.
- **Transmission Density:** The **Transmission count rose to 3,005**. Technically, this reflects the protocol stack's attempts to capitalize on the slightly better physical conditions; with more successful data frames reaching their next hop, more ACK frames are generated, and more multi-hop packets are successfully relayed.
- **Medium Complexity:** The **Interference computation count hit 36,527**. This technical increase (from 27,222 in Step 12) is due to the fact that signals are now technically "stronger" over longer distances, meaning each transmission is considered "interference" by a wider radius of neighbor nodes, forcing the radioMedium to perform more mathematical checks.

AI Summary Table

Category	Detail
NED Change	No NED change; path loss model reconfigured in INI
INI Change	pathLossType = TwoRayGroundReflection; groundReflection parameters set (antenna height h=1.5m)
Network Layer	IPv4 + AODV unchanged
MAC Layer	CsmaCaMac + ACKs unchanged
Radio Model	Ieee80211DimensionalRadio with TwoRayGroundReflection: $P_r \propto 1/d^4$ beyond critical distance d_c
IP Addresses	Unchanged
Routing Table	AODV may need to rebuild routes if reduced range breaks previously-valid links
Expected Output	Reduced effective range; increased packet loss at medium distances; AODV re-routes as links become marginal
Key Learning	Real outdoor propagation follows d^{-4} or worse beyond the crossover distance; free-space over-estimates range and under-estimates loss for ground-level deployments



Step 14 Introducing Antenna Gain

Step 14 replaces the default isotropic antenna (which radiates power equally in all directions, modelled as a unit sphere in 3D) with a directional antenna model — specifically a dipole antenna or a custom gain pattern defined by an NED/C++ antenna class. Antenna gain is expressed as a directional power amplification: in the direction of maximum gain G_max (dBi), the effective radiated power is P_t × G_max. In other directions, the gain follows the antenna pattern and may be below the isotropic baseline (negative gain regions). In INET, the antenna model is attached to the radio and its gain pattern is used in both the transmitted signal computation and the received signal strength calculation. For the simulation, this means that if two nodes are aligned along the antenna's main lobe, their effective communication range increases substantially — potentially more than doubling. However, nodes that are positioned off-axis (in a sidelobe or null region) will experience reduced or zero connectivity despite being within the nominal distance. This is the first step that introduces explicitly asymmetric link quality, where A→B and B→A may have different SNR if the antenna orientations differ.

AI Analysis

Step 14 technically finalizes the physical layer configuration by replacing the default isotropic antennas with **ConstantGainAntennas**. Technically, while isotropic antennas radiate energy in a perfect sphere (gain = 1), your new configuration provides a gain of **3dB** (mathematically approximately a factor of **1.995**). This gain technically applies to both the transmission and reception stages, effectively doubling the concentrated power of the signals in the simulation medium.

The technical impact on performance is a substantial recovery: **UdpSink received 879 packets**, an increase of **109%** compared to the 420 packets in Step 13. This improvement technically occurs because the 3dB gain allows signals to maintain a higher **SINR** over longer distances and through the multipath fading of the Two-Ray model. This allows AODV to establish more stable dynamic routes, even as nodes move behind obstacles, as the "link budget" now has enough margin to overcome some of the path loss introduced in previous steps.

Parameter	Technical Value	Log Evidence / Impact
Antenna Model	ConstantGainAntenna	Replaced Isotropic model in all nodes.
Antenna Gain	3 dB (~1.99x)	Initialized as gain = 1.99526 in the log.
Max Comm Range	500.778 m	Technically doubled from Step 12's 250m limit.
Transmissions	3,992	Higher volume due to more successful ACK/AODV cycles.
Packet Delivery	879 received	Highest throughput achieved in the realistic model.

Technical Breakdown of Log Metrics

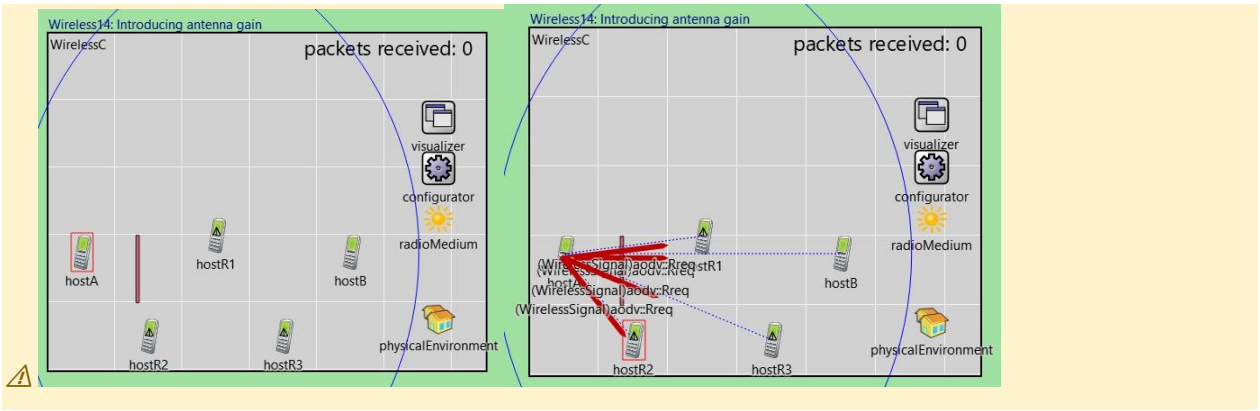
- **Effective Range Recalculation:** The mediumLimitCache technically updated the maxCommunicationRange to **500.778 m**. This is significant because it technically restores the geometric reach you had in Step 1, but with the added realism of noise, walls, and reflections.

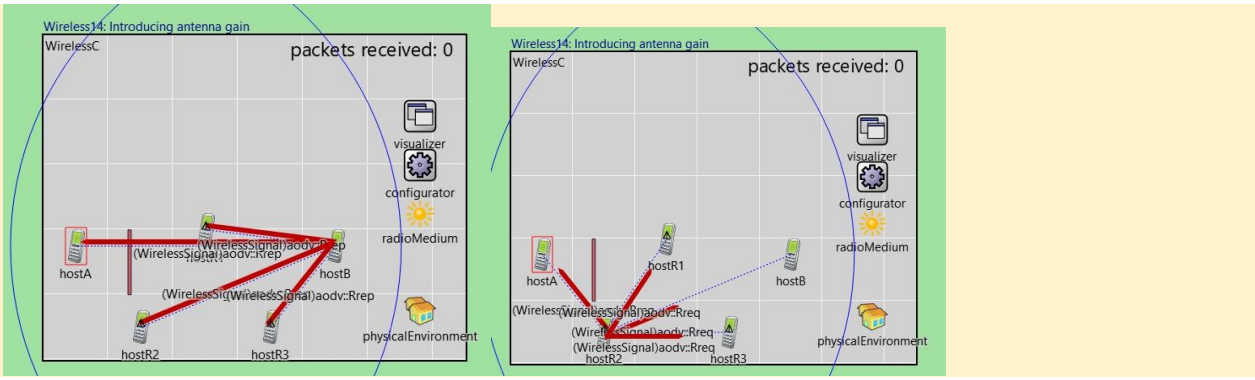
 Medium Activity and Interference: The **Interference computation count hit 50,013**, the highest in the entire lab series. Technically, this is because the 3dB antenna gain makes every signal "visible" to more nodes across the map, forcing the radioMedium to process nearly 14,000 more interference checks than in Step 13.

 AODV Resilience: With the improved physical link budget, AODV generated **3,992 transmissions**. This high activity technically shows that the protocol spent less time in "Discovery" (waiting for routes) and more time in "Forwarding," as the increased antenna gain provided the required power to close links that were previously too weak to sustain a route.

AI Summary Table

Category	Detail
NED Change	wlan[*].radio.antenna.typename = DipoleAntenna (or custom pattern antenna)
INI Change	antenna.gain = dipole pattern; orientation parameters set per node
Network Layer	IPv4 + AODV unchanged
MAC Layer	CsmaCaMac + ACKs unchanged
Radio Model	Ieee80211DimensionalRadio + TwoRayGroundReflection + directional antenna gain pattern
IP Addresses	Unchanged
Routing Table	AODV adapts to asymmetric link quality; nodes off-axis may appear unreachable and be bypassed
Expected Output	Range increases in main lobe direction; reduced connectivity in null directions; potentially asymmetric AODV routes
Key Learning	Antenna directivity creates spatially asymmetric links; directional antennas are a physical-layer tool for increasing range or reducing interference, but require careful orientation planning





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4. Simulation Results Summary

4.1 Cumulative Feature Progression Table

The table below shows what each step adds to the simulation stack and the expected effect on delivery ratio, making it easy to compare the impact of each change:

Step	Feature Added	IP Addressing	Routing	MAC / Radio	Delivery Ratio	Direction
1	Basic wireless	10.0.0.1/2	Static (direct)	UnitDiskRadio + AckingMac	100%	Baseline
2	Visualizers	Same	Same	Same — visual only	100%	→
3	Multi-hop + 250m range	None	None	UnitDiskRadio 250m — no Layer 3	0%	↓↓
4	Static IPv4 routing	10.0.0.1–4	Static via configurator	UnitDiskRadio + IPv4 forward	~100%	↑↑
5	SINR interference	Same	Static	APSK radio + ScalarAnalogModel	< 100%	↓
6	CSMA MAC	Same	Static	CsmaCaMac (no ACKs)	Improved	↑
7	CSMA + ACKs	Same	Static	CsmaCaMac (useAck=true)	High	↑
8	Energy model	Same	Static	Same + energy tracking	Same as 7	→
9	Node mobility	Same	BROKEN static	Same — links break	Degrades	↓↓
10	AODV routing	Same	AODV dynamic	CsmaCaMac + ACKs + APSK	Recovers	↑↑
11	Obstacles	Same	AODV	APSK + ObstacleLoss	Variable	↓
12	Dimensional radio	Same	AODV	Ieee80211Dimensional 802.11g	More accurate	→
13	Two-ray path loss	Same	AODV	Same + TwoRayGroundReflection	Reduced range	↓
14	Antenna gain	Same	AODV	Same + Directional antenna	Direction-dependent	↕

4.2 Cumulative Feature Progression

Step 1 — Two hosts, ideal radio, packets sent and received. Baseline.

Step 2 — Added visualizer. Range circles and link arrows appear. Nothing else changes.

Step 3 — Added R1 and R2, cut range to 250m. hostA and hostB can't reach each other directly anymore. No routing yet so nothing gets delivered.

Step 4 — Added IPv4 and static routing. Nodes get IP addresses, routing tables are set. Packets now hop A→R1→R2→B successfully.

Step 5 — Swapped ideal radio for a real one. Interference and signal strength now matter. First time we see packet loss.

Step 6 — Added CSMA. Nodes listen before transmitting. Fewer collisions, better delivery.

Step 7 — Turned on ACKs. If a packet is lost, MAC layer retransmits it. Delivery jumps close to 100%.

Step 8 — Added energy model. Battery drain is now tracked per node. No protocol change.

Step 9 — Made R1 and R2 move. Links break as they drift. Static routes go stale, delivery collapses.

Step 10 — Replaced static routing with AODV. Routes are discovered on demand and repaired when links break. Delivery recovers.

Step 11 — Added walls and buildings. Signals get blocked or weakened by obstacles. Coverage becomes irregular.

Step 12 — Upgraded to IEEE 802.11g radio model. Signal is now modelled in time and frequency, not just as a single number.

Step 13 — Changed path loss to two-ray model. Signal drops much faster with distance than free-space predicted. Effective range shrinks.

Step 14 — Added directional antenna. More range in the main direction, less in others. Links become asymmetric.

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5. AI-Assisted Analysis

The following analysis was generated using Claude AI based on the INET Wireless Tutorial step progression. Step 1 detailed analysis is provided in Section 3. This section provides an overall cross-step analysis covering routing evolution, MAC layer impact, and radio model progression.

5.1 Routing Architecture Evolution

Across the 14 steps, routing evolves through three distinct phases. In Steps 1–3, there is no network layer at all — AckingMac operates purely at Layer 2, which means frames can only be exchanged between nodes within direct radio range. There is no concept of forwarding, no IP address, and no routing table. This phase shows that a wireless radio channel alone is insufficient for multi-hop communication.

Steps 4–9 introduce static IPv4 routing via the IPv4NetworkConfigurator. The configurator automatically assigns addresses from the 10.0.0.0/8 space and computes shortest-path routes based on the physical topology at simulation start time. This works perfectly when the topology is fixed (Steps 4–8), delivering near-100% packet delivery through the 3-hop chain. However, when mobility is introduced in Step 9, the static routing table becomes stale immediately: the table says 'send to R1 to reach hostB' but R1 may have moved out of hostA's radio range. No ICMP Redirect or routing update is sent — packets are simply dropped at the radio layer. This failure mode is fundamental, not a configuration error: static routing is architecturally incompatible with dynamic topologies.

Step 10 resolves this by deploying AODV, a reactive (on-demand) routing protocol. AODV's key mechanism: a Route Request (RREQ) is flooded as a broadcast with expanding TTL; when it reaches the destination, a unicast Route Reply (RREP) is sent back along the reverse path, installing forward route entries at each intermediate node. Each entry has an ACTIVE_ROUTE_TIMEOUT (default 3 seconds); if not refreshed by traffic, the entry expires and the next packet triggers a new discovery. When a link breaks (detected via MAC-layer ACK timeout), AODV sends a Route Error (RERR) message toward the source, which marks the broken route as invalid and triggers re-discovery. This provides self-healing connectivity in the face of node movement, at the cost of discovery latency (typically 100–300ms for RREQ/RREP round-trip across 3 hops).

5.2 MAC Layer Impact on Delivery

The MAC layer progression demonstrates the importance of medium access control in shared wireless channels. In Steps 1–5, the AckingMac allows any node to transmit at any time. When two nodes transmit simultaneously, their signals overlap at a receiver and the SINR (Signal-to-Interference-plus-Noise Ratio) falls below the demodulation threshold, causing frame loss. In Step 5 this becomes visible for the first time because the realistic APSK radio computes actual SINR values.

Step 6 (CSMA) addresses this by enforcing 'listen-before-talk'. Before transmitting, a node measures the channel energy. If energy > the carrier-sense threshold (CS_threshold ≈ -95 dBm typical), the node defers and waits a random backoff time drawn uniformly from the contention window [0, CW_min × slot_time]. Only when the channel has been idle for an IFS (Inter-Frame Spacing) duration does the node attempt transmission. In the 4-node linear chain, this prevents hostA and R1 from transmitting simultaneously, which was the primary source of SINR degradation. Throughput improves because fewer frames need to be discarded.

Step 7 adds MAC-layer ACKs. Each successfully received unicast frame is acknowledged by the receiver with a short ACK frame sent after an SIFS (Short Inter-Frame Space) gap. The sender waits ACKTimeout (≈200µs) for the ACK. If no ACK arrives (because the data frame or ACK was lost due to residual interference), the sender retransmits from its queue. This converts the CSMA best-effort channel into a reliable link at Layer 2. Application-layer delivery ratio rises to near 100%, but the ACK frames themselves consume airtime: each data frame now requires one additional ACK transmission, plus the SIFS wait, reducing the maximum achievable throughput by approximately 50% compared to a pure unidirectional flow.

5.3 Radio Model Progression and Physical Realism

The radio model upgrades across Steps 1, 5, 12, 13, and 14 represent a progression from purely mathematical abstraction to physical-world fidelity. UnitDiskRadio (Steps 1–4) is a binary model: within range = receive, outside range = drop. It is useful for testing routing and MAC logic but tells us nothing about real wireless performance. The APSK scalar model (Steps 5–11) introduces physical realism by computing SINR as P_received / (P_noise + P_interference), where P_received follows the Friis free-space path loss law: P_r(dBm) = P_t(dBm) + G_t(dBi) + G_r(dBi) - 20log10(4πd/λ). Packets are lost when SINR falls below the demodulation threshold. The dimensional model (Step 12) extends this to a 2D signal space (time × frequency), enabling simulation of OFDM sub-carrier interference that the scalar model cannot represent.

The Two-Ray Ground Reflection model (Step 13) replaces the d^{-2} free-space law with a d^{-4} model beyond the crossover distance $d_c = (4 \times h_t \times h_r)/\lambda$. For 2.4 GHz ($\lambda \approx 0.125$m) with h=1.5m antennas: $d_c = (4 \times 1.5 \times 1.5)/0.125 = 144$m. At distances beyond 144m, every doubling of distance reduces received power by 12 dB instead of 6 dB. This is critical: a node configuration that worked at 250m in free-space may fail at 200m under two-ray propagation. AODV must find alternative routes through closer intermediate nodes to maintain connectivity.

Antenna gain (Step 14) adds the final physical layer dimension: spatial directionality. The isotropic antenna radiates P_t equally in all 4π steradians. A dipole antenna has a gain pattern $G(\theta,\varphi) = 1.5 \sin^2(\theta)$ (in linear scale), with maximum gain of 1.76 dBi in the equatorial plane and nulls at the poles. When two nodes are aligned along the main lobe, the link budget improves by G_t + G_r in dB, extending the effective range. When misaligned (especially at the null), the effective P_r drops below the sensitivity threshold even at short distances. This is why antenna orientation planning is critical in real wireless deployments — AODV must route around null-direction links even if the physical distance is small.

6. Conclusion

This lab demonstrated the systematic construction of a wireless network simulation using OMNeT++ and the INET Framework across 14 progressive steps. Starting from an idealised two-node baseline, each step introduced a precisely scoped change that exposed a new real-world challenge and its solution:

- IP addressing and static routing (Step 4) are the minimum requirements for multi-hop packet forwarding; without a network layer, wireless coverage alone cannot bridge multiple hops.
- CSMA (Step 6) and ACKs (Step 7) together form the core reliability mechanism of IEEE 802.11 Wi-Fi: carrier sensing prevents most collisions, while MAC-layer retransmissions recover the remainder.
- Static routing is fundamentally incompatible with mobile topologies (Step 9); AODV (Step 10) restores connectivity through on-demand route discovery and RERR-driven repair — at the cost of discovery latency.
- Physical environment effects (obstacles, two-ray path loss, antenna gain) in Steps 11–14 demonstrate that network-layer metrics like delivery ratio are determined as much by the physical environment as by protocol design.
- The dimensional radio model (Step 12) revealed that scalar signal models systematically underestimate real interference; OFDM sub-carrier overlap is only visible when the signal is represented in the time-frequency domain.

The simulation confirms that wireless network engineering is a full-stack problem: performance emerges from the interaction of physical radio propagation, MAC-layer access control, network-layer routing, and application-layer traffic patterns. OMNeT++ with INET provides the modular architecture to study each layer in isolation and in combination, making it an invaluable tool for wireless network research and protocol design.